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EDU303 Midterm Preparation

Short Notes By

PIN2 and MUHAMMAD

(MAS All Rounder)

(EDU303)

LESSON 01

INTRODUCTION TO CHILD DEVELOPMENT

Overview of Growth and Maturation

Overview of the Lesson:

What do we understand about the field of child development?

What is developing and growing?

What are individual differences within a normative range of development?

Child Development

Child development is a field that studies the growth and development of the child. It is studied by many different kinds of experts, which include *educationists, teachers, philosophers, psychologists, sociologists, neuroscientists, and social workers*, etc. These experts work independently or together and try to understand the nature of child development. The goal is to understand, and provide insights to problems during development the child is going through.

Why Study Child Development?

Perhaps the most important reason to study child development is to understand how growth and development takes place in the child. This is a basic science question that satisfies the human curiosity; however there are practical benefits of doing so. Who benefits from this understanding? We understand that parents, teachers, educational experts, policy makers, and community in general benefits from this understanding. Experts that study child development address both basic and applied science questions, which are important for the study of development.

Critical Question

Question: In what ways is a child different to an adult, and in what ways the two are still the same?

Continuous or Staged Development

Experts in the field of child development ask if this change in development is continuous or discontinuous, i.e., broken down into stages? The answer lies in the fact how experts view this developmental process.

Naming Developmental Periods

For most of the experts in the field of *Child Development*, an individual growth may be continuous, but when studied it is broken down into stages. The concept of stages of development is not new and has been proposed by ancient thinkers, philosophers, theologians etc. Experts in the field of child development talk about growth and development both in general qualitative and quantitative terms. Many suggest that the individual changes qualitatively over time, backed by measurable quantitative changes. For example, children express lower levels of cognitive functioning when young compared to when they are older. Piaget and Inhelder (1969) not only observed this change

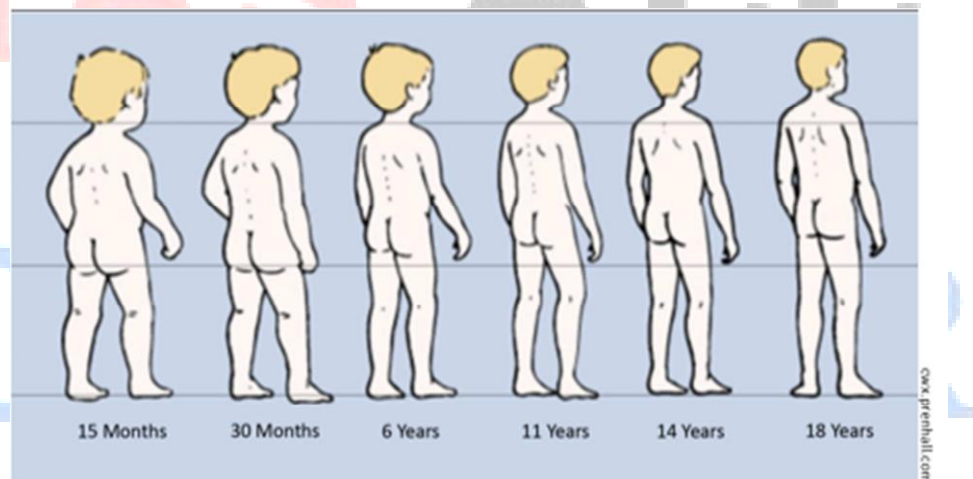
in cognitive development but also were able to quantitatively document it with empirical experiments.

Developmental Periods

Experts divide periods of development into broad and smaller stages and classify them for easy use. When stages, where developmental change is minimal, are close development seems to be more continuous.

What is Developing?

- A. Physical Development. Physical development involves the physical growth and development of the individual starting in the womb, through prenatal period all the way to late adolescence. Some of the changes that we see are: Allometric Development. Body proportions change over development. An infant 15 months old has a bigger head size compared to the body than an adult. The bodily proportions change over the developmental period (see figure below).



Brain Size: In the same fashion the brain also increases in size over development. By six years, its size is very close to what the adult brain is (see figure below).



- B. Cognitive Development: Biological maturation (like age) and, experience (like schooling) are needed together to develop the child intellectually or cognitively. Piaget and many of his colleagues worked on studying cognitive development in the growing child. Piaget went on to say that all cognitive structures are affected by an intertwined relationship between biological maturation and experience, and changed through many stages of cognitive development, like the sensorimotor stage, preoperational stage etc., (Piaget & Inhelder, 1969/2000).
- C. Intellectual Development: Does intelligence change with age? Or experience like going to school improves intelligence? Cahan and Cohen (1989) proposed both; however age had lower effect on intelligence than schooling.
- D. Language Development: Children's vocabulary enriches rapidly. From about 11,000 words in 1st grade to about 80,000 words in 12th grade. In the same fashion grammar in children improves over time.
- E. Emotional Development: Emotions like anger, fear, and happiness can be seen in the infant in a few months, however sadness develops after infancy. From day 1, infants are aware of emotions of others. They respond to other's distress with their own distress. By about 1 year infants can draw emotional cues from others.
- F. Social Development: Early social bonding comes about between infants and their mother. Later siblings and fathers start socially bonding with infants. In Western societies, there are fewer caregivers so infants and children have fewer social relations. This changes when children become older, and develop many social bonds. Healthy social development is based on reliable attachment the primary caregiver (Erickson, 1950; Bowlby, 1969).
- G. Moral Development. Moral development in children occurs along with cognitive development. Around late childhood (8-10 years) children start differentiating between good and bad; although they may be driven by reward and punishment to do that. Conventional moral sense develops much later. All morality according to Kohlberg is universal. However, Kohlberg ideas have been challenged. Jensen argues morality is culturally driven.

Individual Differences:

Most individuals develop normally and follow a typical developmental trajectory. However, during development some individuals are different than others, for example differences in intelligence, achievement, and personality may be obvious among many individuals (which will be discussed in subsequent chapters). Problems associated with development, and thus differences.

Summary:

We have looked at the field of child development. It is important to many experts and laymen alike. The developmental processes are continuous and discontinuous depending on how you see them. Different developments in the individual; physical, cognitive, social, emotional and moral take place over time. Individual differences and similarities in many of the above aspects occur in the growing individuals.

References

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LESSON 02

INTRODUCTION TO CHILD DEVELOPMENT

Factors affecting Development: Biology and Environment

Overview of the Lesson:

What are genes and how does human reproduction takes place?

How traits and genetic abnormalities are inherited?

How genes and environments interact?

How do we measure heritability of traits by Behavior Genetics?

Critical Questions:

How do our genes affect our traits, and our behaviors?

Why would some aspects of our lives be judged by genetic makeup than others?

Genes & Human Reproduction:

Most human cells contain 46 *chromosomes* (23 pairs) that are made from deoxyribonucleic acid (DNA) molecules. The *DNA* consists of two strands of sugar and phosphate molecules like a spiral staircase, where the stairs are made out of nucleotide bases, i.e., adenine (A), thymine (T), cytosine (C) and guanine (G). These nucleotides become the alphabets or building blocks of our genetic code.

A *Gene* is a segment of DNA (base pairs) that provides instructions for a particular structure, function, or trait. So this trait can be like the color of the eyes, or instructions on how to build the membrane of a cell wall (structure) or instructions for moving (function) chemicals like neurotransmitters in neurons from the cell body to its terminals. There are roughly 20-25 thousand genes in human beings.

Human Genome Project:

Based on extensive research, geneticists were able to identify the genetic map of a human. *Human Genome Project (HGP)* completed in 2003 (updated in 2006). Project HGP tells us that there are 3 billion base pairs in human chromosomes. With this project we will be able to identify genetic markers of disease, improving health and prolonging life. Gene therapy will be one such way to help diseased individuals. Parents today can select sex of the child, and can make informed decisions about diseased fetus.

Human Reproduction & Cell Division:

To understand the nature of genes and their effect on living individuals, we must understand a number of biological processes. Human and other life forms begin with the process of *fertilization*, the union of the father's sperm cell with the mother's egg. The fused cell is called a *zygote* and starts to divide and grow into a ball like structure called the *blastocyst* (see figures below).



The sperm and egg (gametes) result from a process called *meiosis*, in which reduction division takes place as opposed to *mitosis*, which involves copy division. After the zygote forms and begins to divide grow, the process of cell division is largely mitosis, however very quickly the dividing cells become specialized forming many different structures in the embryonic and later stages of development. If a zygote divides into two cells and breaks to form two independent zygotes, we have *monozygotic (MZ)* or identical twins as opposed to *dizygotic (DZ)* or fraternal twins. Fraternal twins form through independent fertilization of two sperm and egg cells separately.

Traits & Genetic Abnormalities:

Dominant-Recessive traits are expressed through recessive and dominant genes (alleles) that come from fathers and mothers. Many dominant or recessive gene diseases are due to dominant or recessive

alleles. One such dominant gene disease is *Huntington's disease*. Other recessive gene diseases are *cystic fibrosis*, *sickle-cell disease* and *Tay-Sachs disease*.

Huntington's Disease:

Huntington's disease is a nervous system disease in which some parts of the brain (nerve cells) degenerate. Patients express behavioral disturbances, hallucinations, irritability, moodiness, restlessness or paranoia; it includes quick, sudden, sometimes wild jerking movements of many body parts, the movements can be slow, uncontrolled with an unsteady gait. In addition, dementia, disorientation, confusion, loss of judgment, loss of memory, personality and speech changes may occur. This disease is caused by a genetic defect on *chromosome 4*, due to a dominant allele. The defect causes a part of DNA, to repeat CAG nucleotides many more times than it is supposed to.

Cystic Fibrosis:

Cystic fibrosis is a lung disease, in which thick sticky mucus builds up in lungs, digestive tract and other areas of the body. This is a genetic disease and is a recessive gene disease. This sticky mucus in the lungs can be lifethreatening due to infections and also seriously affects digestive and reproductive systems. It is caused by recessive alleles on *chromosome 7* both from the mother and the father.

Sex-linked Diseases:

X-linked (sex-linked) Traits: A number of genetic traits are linked to alleles on X and Y chromosomes of the mother and the father. Sex-linked diseases are usually found in males and include diseases like *hemophilia*, *muscular dystrophy*, *color blindness* and *retinitis pigmentosa* (a major form of blindness).

- A. Hemophilia. People with hemophilia lack a clotting agent in their blood. The main symptom of hemophilia is bleeding and may occur during surgery or after trauma. Serious bleeding may occur without any cause, including internal bleeding from organs or joints. When males inherit recessive hemophilia allele on X chromosome they cannot mask it with a dominant allele on their Y chromosome.

Chromosomal Disorders:

Genetic diseases can be caused genetic anomalies in chromosomes, or faulty, missing or added chromosomes, e.g., like *Turner syndrome* in which a chromosome is missing, or by presence of an extra chromosome such as in *Down's* and *Klinefelter syndromes*. These diseases account for the majority of miscarriages in women and perhaps are a nature's way of "weeding out" serious defects.

Down Syndrome:

Down's syndrome is marked by a condition of being a "baby", and is marked by maturational retardation. This syndrome causes the individual to have low IQ, facial defects, heart problems, and

shortened life span. This genetic disorder is caused when there is an extra chromosome (chromosome 21) is present.

Prenatal Screening & Genetic Testing:

- A. Ultrasonography (ultrasound). The growing fetus can be monitored for healthy growth and possible disease through a number of screening procedures. Ultrasound uses sound waves. Presents images of the fetus inside the mother's womb and is used to help monitor fetal growth and detect defects.
- B. Amniocentesis. A needle is inserted in the mother's abdomen and cells are withdrawn from the amniotic fluid. Procedure used to detect chromosomal and genetic abnormalities in the fetus.
- C. Chorionic Villus Sampling (CVS). In CVS a catheter is inserted into uterus and cells are taken to test the health of the fetus.

Genes & Environment Interact:

In all individuals, genes and environmental factors are intertwined. These two interact on the development of traits and characteristics of a person in at least four ways. All of the traits are based on genes or their *genotypes*, like bone density in an individual. When a genetic trait is observable, we call it the *phenotype* of that trait, like the color of skin. Genotypes may or may not be manifested as phenotypes.

- A. Range of Reaction. The range of possible phenotypes that exist for a particular genotype.
- B. Canalization. Genetic limits on the effects of the environment. In experiential canalization, in contrast, it is the environment that limits the expression of genes.
- C. Niche-Picking. The tendency to pick activities and environments that fit with our genetic predispositions.
- D. Probabilistic Epigenesis. The likelihood that specific environmental conditions will activate specific genes that will lead to specific outcomes.

Behavior Genetics:

How different are you from your siblings and parents? What accounts for the differences and similarities? Behavior genetics is a field that tries to answer these questions.

Shared & Nonshared Environments:

Individuals with same genetic make, like identical twins, share their *heritability*, an estimate of the degree of genetic influence for a given trait or behavior. Now identical twins can share the same environment. *Shared environments*, lead to experiences that are common across all individuals who are living together. On the other hand *nonshared environment* exposes individuals to experiences that differ across people. One can imagine identical twins living apart in two different environments. Any difference in the degree of a trait that we observe may be attributed to environmental factors in which they are raised.

How Is Heritability Estimated?

There are two ways through which we can estimate heritability across individuals.

1. **Twin Studies:** Comparisons between measurements of identical and fraternal twins, used to estimate the genetic contribution to traits and characteristics.
2. **Adoption Studies:** Comparisons between measurements of children and their adoptive and biological parents used to estimate the genetic contribution to traits and characteristics

Heritability of Complex Characteristics:

1. **Heritability of Cognitive Skills.** Probably the most extensively studied behavioral trait is intelligence. A variety of other cognitive skills (verbal and spatial ability) have a significant hereditary component.
2. **Heritability of Personality and Temperament.** Genetics is an important influence on personality and temperament.

Summary

In this topic we have looked at chromosomes, DNA and genes, and the Human Genome Project (HGP). We also reviewed human reproduction, traits & genetic abnormalities, and processes of prenatal screening and genetic testing. Finally we talked about interactions between genes and environment.

LESSON 03

INTRODUCTION TO CHILD DEVELOPMENT

Twin and Adoption Studies

Overview of the Lesson:

Look at the field of behavior genetics.

Talk about heritability measurement through heritability estimates and concordance rates. Discuss heredity and environment interaction.

Critical Questions:

How can we use kinship and adoptions studies to understand development in children?

What do these studies tell us about the effects of heredity and environment on development?

Behavioral Genetics:

Behavioral genetics (especially human) seeks to understand both the genetic and environmental factors that lead to individual differences in human behavior. Sir Francis Galton (1822-1911) was the first scientist to study heredity and human behavior systematically, with respect to intelligence.

Galton and Intelligence:

Galton was interested in the fact whether human abilities like intelligence were inherited. He reasoned that intelligence should be similar among relatives than general population. Second and third degree relatives he found were indeed less similar in intelligence compared to first generation (siblings).

Methodology:

For Galton, studying kinsmen was one way of determining the effect of heredity on human traits, like intelligence. However the efficacy of this methodology to relate heredity to human traits could not be as effective as studying members of family who are genetically alike, like identical twins. Identical twins share genetic information identically. Thus provide a window to study heredity effects on humans' traits.

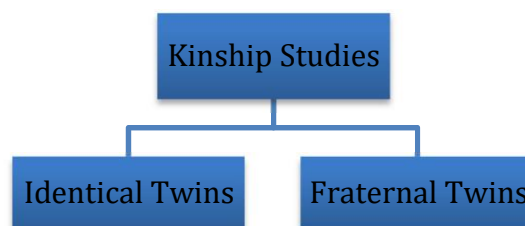
Questions to Consider

Some behavior geneticists think it is possible and useful to answer *how much* heredity factors contribute to traits. However, there is now a growing consensus among experts who believe that question above is unanswerable and perhaps not very useful. These experts believe heredity and environment are inseparable and thus work together. Let us look at both approaches one by one.

Heritability Estimates:

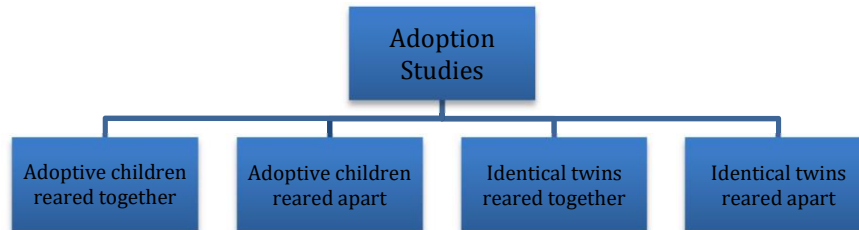
Heritability estimates measure individual differences in complex traits in a specific population that are due to genetic factors. Heritability estimates are obtained from kinship studies (twin studies), which compare the characteristics of family members. Modern experts use genetically related twins (identical and fraternal) to measure these estimates that range between 0.00 and 1.00.

Kinship Studies:



Kinship studies (raised by biological parents) provide experts to look at gradient of genetic factors on inherited traits. Identical twins are genetically 100% similar and fraternal twins much less. And since they are raised together, a trait, say depression, if present in one of identical twins, would also be (highly likely) present in the other twin. In fraternal twins sharing of this (depressive) trait would be much lower (less likely).

Adoption Studies



Likewise adoptive children (fraternal twins or siblings) reared together will bring forth environmental effects on their traits when compared to adoptive children reared apart.

A. Cognitive Skills. Heredity has a low to moderate influence on IQ scores (estimates at .50 heritability). Similar findings for a variety of specific cognitive skills i.e., spatial reasoning, verbal reasoning, and perceptual speed have been elucidated. Also similar data has been collected for achievement scores i.e., English usage, mathematics, social studies, and natural sciences. B. IQ Similarity Among Relatives

Personality and Temperament:

Heritability estimates of personality and temperament go as high as .50, and change over time, which are attributed to changing environment. Heritability estimates for hyperactivity, schizophrenia, obesity, and alcoholism are fairly high.

Personality Traits:

If we look at NEO-PI five personality traits, identical twins share these traits to a greater extent than fraternal twins.

Concordance Rates:

Heritability of traits is also measured by concordance rates. Concordance rates determine what percent do both twins show a trait, when it is present in one twin. These measurements range from 0 to 100%. A score of 0 means if one twin has a trait the other twin does not have it. And a score of 100 means that if one twin has a trait the other one also has it.

Limitations:

Experts have raised serious questions about the accuracy of heritability estimates and concordance rates.

1. Each measure refers to particular population, with unique genetic and environmental influence.
2. Accuracy of measures depends upon twin pairs used to reflect genetic and environmental variation.
3. Limited usefulness. These are interesting numbers but do not tell us how traits develop.

Developing Traits: How?

Range of Reaction: Each individual has a unique genetic make and each responds to the same environment differently. Sometimes different genetic-environmental combinations can make two people attain the same level of a trait or ability.

Canalization: Canalization is the tendency of heredity to restrict the development of some characteristics to just one or two outcomes. When we look at behaviors or traits that are highly constrained by heredity, such restrictive canalization has adaptive value. Because of this many children will develop certain specie-typical behaviors that will make them survive wide range of rearing conditions.

Genetic-Environmental Correlation: Genetic-Environmental correlation refers to the fact how genes influence environment in which we live.

1. **Passive correlation:** When the child is young she has no control over his or her environment. Parents provide environments.
2. **Evocative correlation:** When the child can evoke a changes in the environment. Happy baby gets more stimulation.
3. **Active correlation:** At older age, children extend their experiences beyond family choosing environments that compliments their heredity (also called niche picking).

The Epigenetic Framework:

Epigenesis means development resulting from ongoing, bidirectional exchanges between heredity and all levels of our environment.

Summary:

We looked at the field of behavior genetics. We also talked about heritability measurement through heritability estimates and concordance rates. Lastly we discussed heredity and environment interaction.

LESSON 04

INTRODUCTION TO CHILD DEVELOPMENT

Continuity of development

Overview:

Early and continuity of development in the neonate: reflexes, sleep and arousal patterns, crying behavior. Neonatal Behavior Assessment Scale and Sudden Infant Death

Learning and
memory Motor
development

Sensory and perceptual processes, intermodal processing and explanations of intermodal processing

Critical Questions:

What are some of the stages through which the individual passes to early in life?

What are some considerations we need to keep in mind with reference to normal and abnormal development in early life?

Newborn reflexes include eye blink, sucking, swimming, moro, palmar grasp, tonic neck, stepping, Babinski, etc.



Palmar Grasp



Babinsky



Swimming

Infant States of Arousal:

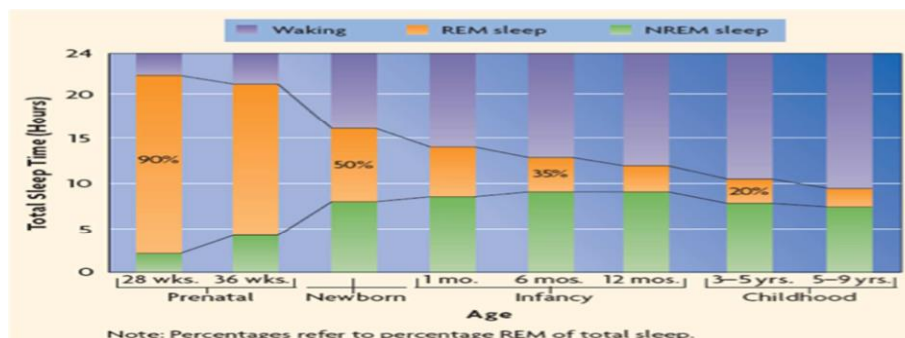
These include regular & irregular sleep cycles, drowsiness, quiet alertness, waking activity and Crying

Sleep Patterns:

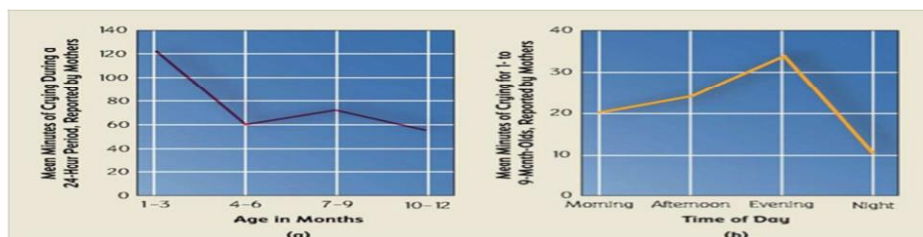
1. Sleep moves to an adult-like night-day schedule during the first year.
2. Need for sleep declines from 18 to 12 hours a day by age 2.

Childhood Sleep Changes:

See the graph below to changes in sleep pattern as the child grows. The graph also shows different kinds of sleep.



Infant Crying Patterns:



Adult response to infant cries depends on interpretation of cry: Adults use cry *intensity* and *context* to judge the nature of cry. Accuracy of this cry improves with experience. Adult can respond to the infant with empathy; develop a child-centered attitude; and think that they have control over the infant crying.

Ways to Soothe a Crying Baby:

- Hold on shoulder and rock or walk
- Swaddle
- Pacifier
- Ride in carriage, car, swing
- Combine methods
- Let cry for short time

Neonatal Behavioral Assessment Scale:

Neonatal Behavioral Assessment Scale (NBAS) was developed by Brazelton (1973) to assess a wide range of behaviors in newborns and infants up to two months old. After assessment it describes baby's strengths, adaptive responses and possible vulnerabilities. It can be used for discovering individual & cultural differences. Helping parents get to know their babies. Predicting development based on changes in scores.

Sudden Infant Death Syndrome (SIDS):

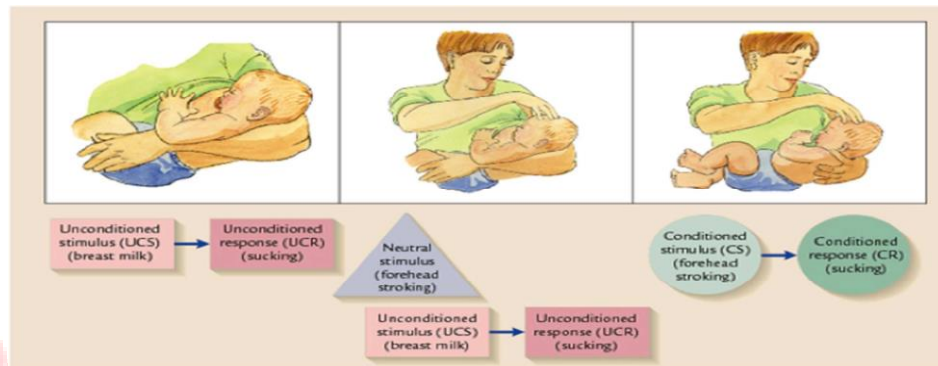
Sudden infant death syndrome (SIDS) is the unexpected, sudden death of a child under age 1 in which an autopsy does not show an explainable cause of death; peaks between 2 and 4 months of age. It is a leading cause of death of infant mortality in industrialized nations. Quitting smoking,

changing an infant's sleeping position and removing a few bedclothes can reduce the incidence of SIDS.

Learning & Memory:

Classical Conditioning:

An infant can be conditioned using classical conditioning processes. Figure below shows how an infant becomes conditioned to sucking response during breast-feeding.



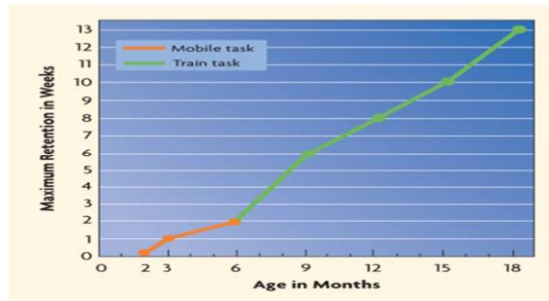
Operant Conditioning:

In the same fashion the infant can be trained to operant conditioning processes by the use of reinforcements and punishments.

Reinforcement	Punishment
<i>Increases</i> probability of behavior	<i>Reduces</i> probability of behavior
Presenting reinforcing stimulus	Presenting punishing stimulus
Removing punishing stimulus	Removing reinforcing stimulus

Infant Memory:

Infants not only learn, but also retain information. This retention of information increases as the infant grows. Two tasks that are age specific are used to assess infant retention.



Habituation to Study Infant Memory:

Other ways to assess infant memory is through the process of habituation to a stimulus. Infants like adults habituate to a stimulus, when shown a new stimulus along with the old, infants display retention for immediate learnt material. Paying attention to old stimulus later on shows retention of material long ago.

Imitation:

Newborns can imitate, but is harder to induce in older babies. Some researchers suggest it is a reflex. Capacity to imitate improves with age, and helps them learn. Neuroscientists believe that imitative behavior is caused by mirror neurons.

Motor Skills Development

Cephalocaudal (Head to tail): Control of head before arms and trunk, which are before legs.

Proximodistal (Center to periphery): Control of head and trunk before arms and legs.

Complexity of Motor Skills:

Increasingly complex *systems* of action develop with each skill. Four factors in each new skill:

1. CNS development
2. Body's movement capacity
3. Child's goals
4. Environmental supports

Steps in Reaching and Grasping

1. Prereaching
2. Reaching with two hands, then one.
3. Ulnar Grasp
 - Adjust grip to object.
 - Move objects from hand to hand.
4. Pincer Grasp
 - Using the thumb and forefinger to pick up things

Some Milestones in Reaching

Infants' Sense of Touch:

Newborns display reflex responses to touch on mouth, palms, soles, genitals and later on to exploratory mouthing. Pleasurable touch releases endorphins in the brain.

The infant is sensitive to pain. Pain can affect later behavior, i.e., long lasting pain can have devastating effects on the growing infant and can have serious psychological and behavioral consequences. Pain can be relieved with anesthetics, sugar, gentle holding.

Taste and Smell:

Infants prefer sweet tastes at birth and quickly learn to like new tastes. Have preferences of odors from birth, and is affected by mother's diet during pregnancy. Can locate odors and identify mother by smell from birth

Sense of Hearing:

Infants can hear a wide variety of sounds at birth and even before birth, especially in the last trimester. They prefer complex sounds to pure tones, and learn sound patterns within days. Infants are sensitive to voices and biologically prepared to learn language.

Developments in Hearing

1. 4 -7 months: Sense of musical phrasing
2. 6 – 8 months: "Screen out" sounds from non-native language, recognize familiar words, natural phrasing in native language.
3. 8 – 9 months: Detect syllables that often occur together in the same word.

Sense of Vision

Vision is the least developed of senses at birth. The infant is unable to see long distances, focus clearly with limited binocular visual ability. With the presence of sounds the infant can scan environment and try to track interesting objects. Color vision improves in first two months.

Improvements in Vision

Brain development helps infants reach adult levels of vision. At 2 months, focus and color vision develops. At 6 months, acuity, scanning & tracking; and around that time (6–7 months) depth perception becomes better.

Stages in Depth Perception

Birth – 1 month	Sensitivity to kinetic cues
2 – 3 months	Sensitivity to binocular cues
5 – 12 months	• Sensitivity to pictorial cues • Wariness of heights

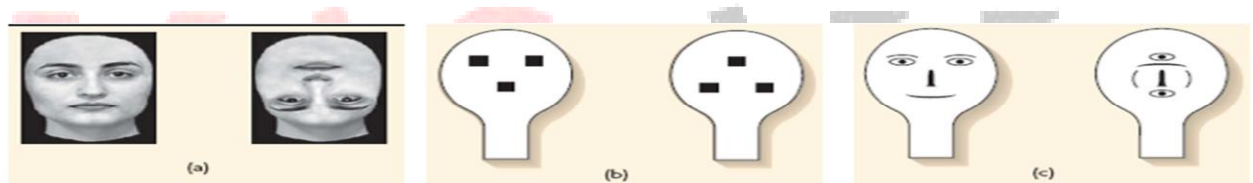


Stages in Pattern Perception:

Early on in an infant's life perception is governed by simple patterns. Infants and young children are unable to perceive fully complex stimulus patterns.

Face Perception:

Newborns prefer simple face-like objects (left in panel b) over the one on the right. This ability goes away in about 6-months. Infants prefer regular face to a scrambled face (panel c).



Size Perception

Infants by 1 week of age have the ability to perceive differences in size.

Object Unity

And by 2 months age they can perceive object unity.

Stages in Intermodal Perception:

1. Birth: Largely amodal sensory perceptions.
2. 3-4 months: Prefer "matching" sights and sounds
3. 5-6 months: Reach for object in the dark, coordinating sight and touch

Differentiation Theory:

Differentiation theory proposes that infants search for *invariant* features of the environment.

And, note *stable relationships* between features, like visual patterns, intermodal relationships, etc. Gradually detect finer and finer features. *Differentiation!*

Affordances

Gibson (1966) proposed that individuals discovered invariant features of the environment by acting on it. The environment “affords” such features and makes actions future-oriented. So for the infant it is the environment that makes it possible to perceive objects and stimuli.

Summary:

We looked at:

1. Early development in the neonate with reference to reflexes, sleep and arousal patterns, crying behavior.
2. Assessment measures like Neonatal Behavioral Assessment Scale; studied learning and memory. And motor development
3. And finally discussed sensory and perceptual processes and intermodal processing and its explanation.

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LESSON 05

RESEARCH METHODS IN CHILD DEVELOPMENT

The National Survey and Epidemiological Studies

Overview:

Population and samples, definitions and use in child development. Kinds and methods of sample extraction, sampling errors and biases Epidemiological studies on children through development. National epidemiological studies on children of Pakistan

Critical Questions:

What are some of the basic research procedures in studying child development?

Why is it important to carry out epidemiological studies to get a better grasp of child development?

Population & Samples:

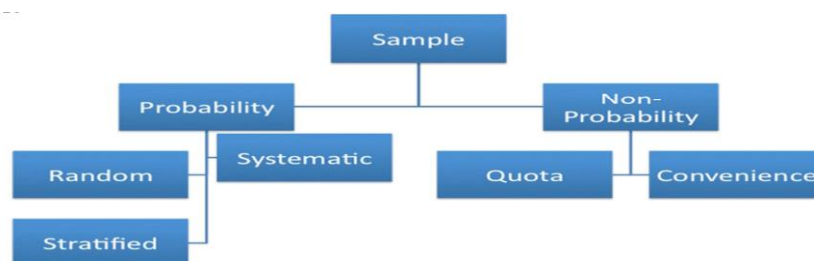
For a majority of areas in behavioral and social sciences research is driven by *statistics* and methodology that incorporates statistical *procedures* and *analysis*. At the very core of these statistical procedures is the concept of population and sample or samples that are drawn out of it.

Mathematically and statistically speaking a population is a large conglomerate of all the members or characteristics of these members in which the researcher is interested, like weight, intelligence, happiness, achievement etc. These characteristics are called *parameters*. Since research cannot be carried out on all members of a population, investigators draw samples out of this population and study these characteristics, now called *statistics*.

Kinds of Sample:

When individuals in a population have an equal chance (greater than zero) of being drawn into a sample we call it *probability sample*. On the other hand if some members of a population have no chance (equal to zero) being drawn into a sample; that sample is called *non-probability sample*.

Varieties of Sample:



Probability Samples:

Random Sample: When each member in a population has an equal probability of being selected. Systematic Sample: Sampling begins by randomly selecting one member, and then selecting every n th member onwards.

Stratified Sample: When population composed of distinct categories, it leads to a sample based on these categories and is called stratified sample.

Non-Probability Samples:

Quota Sample: In quota samples, population's subgroups are proportionately sampled.

Convenience Sample: members of the population are chosen based on their relative ease of access. Sampling friends, co-workers, or shoppers at a store are all examples of convenience or accidental sample.

Sampling Errors:

Selection bias or sampling bias occurs when probabilities of selecting members into a sample differ from true probabilities contained in a population. In many ways non-probability sampling can suffer from such bias and can affect survey research.

Random sampling error: Or estimation error is an error that is caused by observing a sample instead of the whole population.

Epidemiology:

This branch of medicine studies patterns of health (or disease) in a population. Children epidemiological studies look at child diseases that focus on their causes of death (COD) or mortality. Two kinds of mortality have been the center of attention for World Health Organization (WHO).

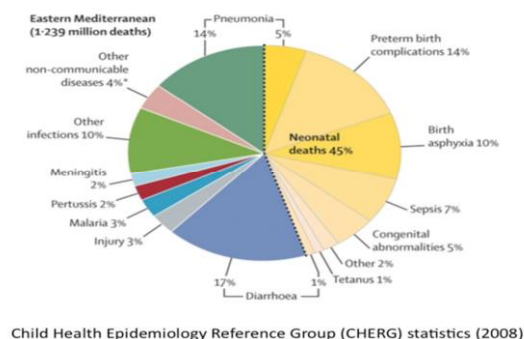
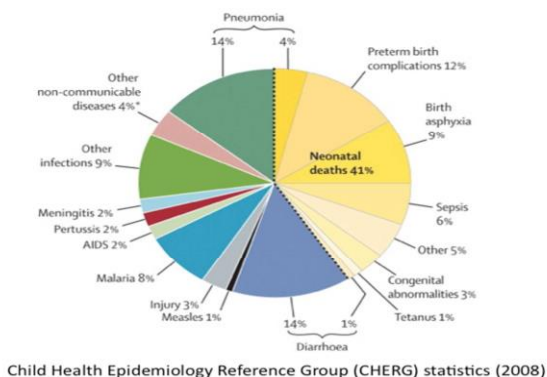
- Neonatal mortality (birth-28 days)
- Child mortality (1-59 months)

Child Mortality:

WHO Statistics tells that [About] 7.6 million children under the age of five die every year, according to 2010 figures. Leading causes of death are pneumonia, diarrhea, malaria and health problems during the first month of life. Over one third of all child deaths are linked to malnutrition according to statistics provided by WHO. Children in low-income countries are nearly 18 times more likely to die before the age of five than children in high-income countries.

Mortality Distribution: Global

Mortality Distribution: E. Mediterranean



Infant & Child Mortality in Pakistan:

Major reasons for neonatal and childhood mortality in Pakistan (Agha, 2000; Hobcraft, McDonald & Rutstein, 1985) includes:

1. Poor child-spacing in pregnancies
2. Water and sanitation issues
3. Social and gender inequities
4. Socio-economic issues

Goals:

Though neonatal and childhood mortality has decreased in Pakistan from 1980s to present times. The goal is to cut this mortality to two-thirds by 2015. Economic and financial burdens in Pakistan may delay these goals.

Summary:

- We looked at population and samples, and they are used in research on child development.
- We also discussed different kinds of samples and their extraction for research purposes. In addition we identified sampling errors and biases.
- We then concentrated on epidemiological studies on children through development and talked about studies conducted for Pakistan.

LESSON 06

RESEARCH METHODS IN CHILD DEVELOPMENT

Self-Reports and Observation

Overview:

Brief outlook on methods used for gathering data about children.

Self-reports, clinical interviews, questionnaires and tests

Observational method and its kinds

Psychophysiological methods

Advantages and disadvantages of the above mentioned methods

Critical Questions:

Why do we use different kinds of methods to study children?

What is the goal of all of these methods?

Why should we learn about these and other research strategies?

Self-Reports:

Interview:

Interview or *clinical interview* is very flexible, conversational styled self-report in which we want to know participant's or child's point of view.

1. Unstructured Interview: These interviews are highly flexible, use open-ended questions and gather as much information about the child as possible.
2. Structured Interview: Each participant is asked same questions in same way. Usually questions in this interview are closed-ended. In structured interview, questionnaires may be used to get answers from groups.

Limitations

Accuracy of participants' expressions may be misjudged. There may be distortions in participants' recall or judgments. Flexibility may make responses too varied to affect accuracy.

Questionnaires:

1. For very young non-school going children self-reporting questionnaires are not possible or fruitful, but older children can respond to questionnaires on nominal or ordinal scales.
2. Many younger children can respond to questions in a dichotomous way, and are easier to read and respond to by young children.
3. With older children who go to primary or secondary schools we can use questionnaires that are based on equal-interval questions.

Tests:

Still other forms of tests can be used to measure children's intelligence or mental disorders. Amongst these one is the draw-a-person test (or draw-a-man test), originally conceived and designed by Goodenough (1926). The test expresses dramatic differences between schizophrenic and normal children (Jaynes, 1976).

In the Draw-a-Person Test the child is asked to draw a man, a woman and him- or her-self. Clinicians carry out analysis on these three drawings.

Observation:

Perhaps the oldest method used by any class of investigator (or thinker) has been observation. Observation method requires observation of phenomena using a variety of sensory modalities. This method has been extensively used to study psychological phenomena in children, adults and animals. There are two kinds of observation:

Naturalistic Observation: In naturalistic or “field” observation, investigator observes behavior where it happens.

Structured Observations: Usually take place in the laboratory situation where behavior of interest can be evoked. All participants have equal chance to display behavior.

Sampling through Observation

Event Sampling: Observer records all instances of a particular behavior (events) during a specific time period.

Time Sampling: Observer records certain behaviors that occur during a time interval.

Observing Events:

Detailed observations of, say classroom learning, leads to classification of various events of the phenomena, e.g., classroom learning can involve verbal, perceptual, motor, problem solving forms of learning.

Appraisal:

Advantages	Disadvantages
Important first step to study any phenomenon (including learning)	Complex and difficult
	Observer's influence: Participants may react in unnatural ways (can be controlled)
	Observer Bias: Record what they expect, rather than what really happens. It is a serious danger in research.
	Uncontrolled

Psychophysiological Methods:

Autonomic Nervous System Activity

Measures of autonomic nervous system activity, which include heart rate (ECG), skin conductance (GSR), blood pressure, respiration, pupil size, stress hormones, etc.

Measuring Brain Functioning:

Functional Magnetic Resonance Imaging (fMRI): Functional Magnetic Resonance Imaging (fMRI) visualizes brain function, by changes in chemical composition of brain areas or changes in the flow of fluids (blood) that occur over time.

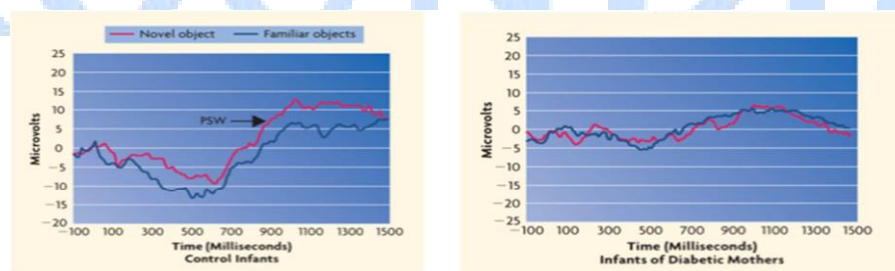
Positron Emission Tomography (PET): Increased radio-labeled glucose activity is scanned in the brain while subjects engage in different cognitive processes.

Structure function relationship, spatial resolution is good while temporal resolution is not good. It is an invasive procedure.

Near-infrared Optical Topography (NIROT): NIROT is a noninvasive procedure that simultaneously acquires hemoglobin absorption from an array of optical fibers on the scalp to construct maps of cortical activity (Kennan et al., 2002).

Event-related Potential: Event-related potential measures the brain's electrical activity (potentials) as it corresponds to impinging stimuli (events). Excellent temporal resolution (faster response) compared to PET or fMRI.

Memory Impairments in Infants: Nelson et al (2000) demonstrated that young infants' memory performance was affected by maternal diabetic load during gestation. Infant's memory deficits resulted from iron depletion in brain areas measured by ERP.



Limitations:

1. Interpreting psychophysiological results require a great deal of inference. Researchers cannot be sure that an infant or a child has processed the information in a particular way.
2. Many factors can influence physiological response, so fatigue, boredom, drive and emotions in a child can affect physiological responses.

Summary:

We looked at self-reports, clinical interviews, questionnaires and tests. We also discussed observational method and its kinds. Finally we talked about psychophysiological methods, and discussed the advantages and disadvantages of these methods.

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LESSON 07

RESEARCH METHODS IN CHILD DEVELOPMENT

Correlation and Experimentation

Overview:

What is correlation method?

How do we make predictions using correlation method?

What is experimental method?

What is hypothesis testing and what is its relationship with theory and research? How does experimental method determine cause-and-effect relationship?

Critical Questions:

Why do we want to use experimental method in child development? How are correlation and experimental methods different?

Science:

Science seeks to generate confirmable propositions (hypotheses) by fitting a formal system of symbols to empirical observations (Stevens, 1951). Hidden in this definition are two important philosophical positions of rationalism and empiricism. Combined, these two positions create a powerful discipline of inquiry.

Scientific Method:

The scientific method involves gathering of data either through observations or manipulations. The investigator first forms hypotheses and tests them to draw conclusions. Conclusions get revised if necessary with additional observations or manipulations. These conclusions form the basis of theories and theory-building.

Two Important Research Methods:

1. Method of Correlation
2. Experimental Method

Correlation:

Research methods that attempt to determine the strength of a relationship between two or more variables or behaviors.

For example, does maternal warmth leads to children compliance?

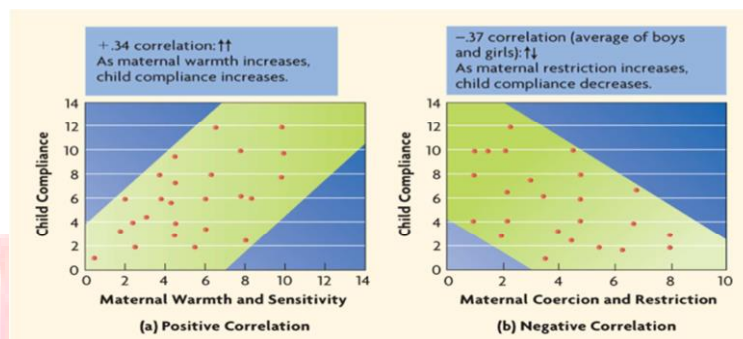


Correlation Coefficients:

Magnitude of correlation is represented as a number between 0 and 1. Closer to 1 shows a stronger relationship than 0.

Direction of correlation is indicated by + or – sign. Positive (+): as one variable increases, so does the other. Negative (-): as one variable increase, the other decreases.

Examples:



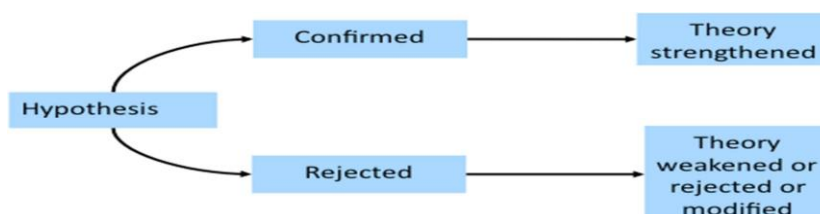
Experimental Method

Experimental method captures the empirical side of science and has many advantages.

Advantages	Disadvantages
Very productive	At times cannot be used for ethical reasons
Controlled	Difficulty of generalization
Repeatable	
Unambiguous	
Objective	

Hypothesis:

Predictions or hypotheses are testable statements that propose a causal relationship between manipulated and measured variables.



Theory:

A theory is a system of interrelated hypotheses that explain a number of observations or phenomena. There are two aspects of theory, namely,

1. Formal aspect of a theory includes words and symbols that explain observations.
2. Empirical aspect of a theory comprise of physical events and observations that are measured and quantified.

Theory & Research:

In the end research evidence (data, empirical aspects) feed a theory (formal aspects). And theory in turn proposes new predictions that need to be conformed to data.



Characteristics of a Theory:

1. Synthesizes observations
2. Generates new empirical hypothesis
3. Confirmed hypothesis makes a theory strong; else theory is modified or dropped
4. Theory is not right or wrong, only useful or not.
5. Theory is chosen based on the law of parsimony.
6. Theories contain formal and empirical aspects
7. Theories must explain empirical events.

Scientific Law:

If a hypothesis is confirmed many a times, it eventually becomes a law, and usually is represented with symbolic (mathematical) notation.

How to Conduct an Experiment?

1. Researchers begin by randomly assigning subjects to either:
 - a. *Experimental group*: receives treatment.
 - b. *Control group*: does not receive treatment.
2. Random assignment is an unbiased procedure that assigns participants to treatment conditions randomly.

Variables in an Experimental Study:

1. *Independent Variables*: Experimenter changes, or manipulates this variable. Expected to cause changes in another variable.
2. *Dependent Variables*: Experimenter measures the affected variable. Expected to be influenced by the independent variable.

Kinds of Experiments:

1. *Field Experiment:* Use rare opportunities for random assignment in natural settings.
2. *Natural Experiments:* Include quasi-experiments. Compare differences in treatment that already exist. Groups chosen to match characteristics as much as possible.

Experimental Methods used in Assessing Child Development:

Longitudinal: Same participants studied repeatedly at different ages.

Cross-sectional: People of different ages all studied at the same time

Sequential: Same groups of different-aged people studied repeatedly as they change ages.

Micro-genetic: Same participants studied repeatedly over a short period as they master a task.

Longitudinal Design:

1. *Advantages:* Permits study of common patterns and individual differences in development and relationships between early and later events and behaviors.
2. *Problems:* Age-related changes may be distorted because of biased sampling, selective attrition, practice effects, and cohort effects. Theoretical and methodological changes in the field can make findings obsolete.

Cross-sectional Design

1. *Advantages:* Efficient, not plagued by selective attrition, practice effects, or theoretical and methodological changes in the field.
2. *Problems:* Does not permit study of individual developmental trends. Age difference may be distorted because of cohort effects.

Sequential Design:

1. *Advantages:* Sequential design permits both longitudinal and cross-sectional comparisons. Reveals cohort effects. Permits tracking of age-related changes more effectively than the longitudinal design.
2. *Problems:* May have the same problems as the longitudinal and cross-sectional strategies, but the design itself helps identify difficulties.

Microgenetic Design:

1. *Advantages:* Offers insights into how change occurs. Requires intensive study of participants' moment-by-moment behaviors.
2. *Problems:* The time required for participants to change is difficult to anticipate. Practice effects may distort developmental trends.

Summary

We looked at what is correlation method is and how do we make predictions using this method. We then discussed what experimental method is and its kinds? What is hypothesis testing, and what is the relationship between theory and research? Lastly we saw how experimental method determines cause-and-effect relationship.

LESSON 08

RESEARCH METHODS IN CHILD DEVELOPMENT

Case Studies and Studies: Using Combined Methodologies

Overview:

What do we understand about case study or case history method? Putting a number of methodologies together to study the growing child We will also briefly look at ethnography to study groups.

Critical Question:

Why do child specialists use a multitude of methodologies to assess child development?

Case Study Method:

1. A case study method is an in-depth analysis of a person like a child, or a group of individuals, or events etc.
2. In many ways the case study method has been used in social and physical sciences, and its use goes back to ancient times.
3. In modern times, Freud (1895) used case study method heavily and made significant strides in using it as a clinical tool.
4. Case study methodology is used in qualitative research, and collects-analyzes descriptive and explanatory data about the case.
5. Case study methodology, unlike cross-sectional sampling in experimental research, samples phenomena on longitudinal or sequential basis.
6. The *subject* (case) provides the instance of the phenomena, and the *object* of the study is to understand the phenomena by using the subject.

Types of Case Studies:

1. Illustrative Case Studies: These are primarily descriptive studies.
2. Exploratory (or pilot) Case Studies: These are performed before implementing a large-scale investigation.
3. Cumulative Case Studies: These serve to aggregate information from several sites collected at different times.
4. Critical Instance Case Studies: These examine situations of unique interest with little to no interest in generalizability.

Multimodal Approach:

Case study method brings together a wide range of information on one child by using:

- Interviews
- Observations
- Test scores
- Psychophysiological measures

Case Study Method: Validity

Like any other psychometric instrument case study method needs to be valid. A method is valid if it measures the concept it is trying to measure.

Case Study Method: Reliability

Reliability of case study means, whatever phenomenon is being studied with this method should bring the same or similar results when repeated.

There are some advantages as well as disadvantages of case study method.

Strengths:

Flexibility: The case study approach is a comparatively flexible method of scientific research.

Emphasis on context: By seeking to understand about a single subject or small group of subjects, case studies specialize in “deep data,” or “thick description”.

Weaknesses:

Inherent subjectivity: The case study is often criticized as being too subjective and even pseudoscientific.

High investment: Case studies can involve learning more about the subjects being tested than most researchers would care to know.

Ethical consideration: Conflict of interest, biases, opinions

Ethnography:

Descriptive, qualitative technique used to study and understand a culture or social group. The method is borrowed from anthropology. In participant observation the researcher lives in community for months or years to observe and participate in the society he or she lives in.

Invisible Beings:

Ethnographic studies have revealed that East Indian Hindu parents encourage their children to communicate with “invisible” characters, whereas Christian fundamentalists discourage that (Taylor & Carlson, 2000).

Following are the strengths of ethnographic studies:

Strengths:

Social desirability: Since the investigator blends with social group, participants’ behaviors are not influenced.

Detailed longitudinal observation: The researcher can gather in-depth understanding of cultural group by staying with them for extended period of time.

Weaknesses:

Subjectivity: Ethnographic studies can be subjective and biased.

Interpretations: The investigator may be affected by his interpretations of the phenomena in questions, than what actually exists.

Summary

We looked at case study or case history method. We discussed that a number of methodologies can be used to study a case or a few cases. We also looked at ethnographic studies that study culture and groups.

THEORIES OF DEVELOPMENT

Structural-organismic Perspectives: Psychodynamic Theory

LESSON 09

Overview:

We will look at the inherent nature of structural-organismic perspective. And then discuss Freud’s theory of psychosexual development.

Critical Questions:

How structural-organismic theories did come about as a natural outcome of explaining growth of the individual, especially the child?

How did Freud conceive of child development?

Structural-Organismic Perspective:

1. The organismic approach uses *root metaphor* to model growing biological organism (Reese & Overton 1970).
2. Growth unfolds an organism towards its adult form (end point).
3. And this structural-organismic perspective carries out a logical analysis of the developmental changes taking place from beginning to end (Kaplan 1983).
4. Structural-organismic approach is thus based on laymen thinking that a developing individual changes in stages. Just like a mango grows from seed to giant tree.
5. Variation at each stage is deemed negligible.



Structural-Organismic Perspective:

1. For Piaget, it is the thought that develops through stages and finally reaches the stage of formal operations.
2. For Freud, it is the sexual development of the individual that takes him through a number of stages finally ending with the genital stage, marked by mature sexual interests.
3. Both perspectives are based on the idea of determinism.

Psychodynamic Perspective:

Freud a neurologist, was attracted to hypnosis, and studied mental patients through clinical interviews.

He developed psychoanalytic theory that explains human personality, development and mental abnormality.

Psychoanalytic Theory:

Freud proposed that *childhood sexuality* and *unconscious motivations* influence *personality*. Human personality according to Freud was composed of *Id*, *Ego* and *Superego*. These personality components exist in *unconscious*, *preconscious* and *conscious* domains.

Model of Personality:

The iceberg model of personality depicts the unconscious (underwater), conscious (above water), and preconscious (at the surface of the water). The *Id* hides underwater in the unconscious mind, and is a reservoir of mostly unacceptable thoughts, wishes, feelings and memories. *Superego*, partially unconscious, represents internalized ideals (the conscience). *Ego*, mediates the demands of id and superego.

Psychosexual Stages:

Psychosexual stages or periods have a characteristic sexual focus that leaves their mark on adult personality.

Foundations of personality were laid down within the first *five* years of life. The five developmental stages are:

- Oral
- Anal
- Phallic
- Latency
- Genital

Oral Stage:

The oral stage of the child spans from birth to 24 months of age. In this stage the infant receives pleasure from his or her mouth (oral zone). Activities such as sucking, chewing and biting deliver pleasure. Fixation at this stage leads to excessive eating or smoking when adult.

Anal Stage

This stage of the child development spans from 2-3 years of age. Pleasure focuses on bowel and bladder elimination. The child has to cope with the demands of control (expulsion and retention). Excessive punishment (usually mother's) to control bowel and bladder elimination can lead to hostility (towards her).

Phallic Stage

In this stage the child's age ranges from 3-7 years of age. Pleasure focuses on the genitals. Boys tend to have erotic feelings towards mother, hostility towards father. Also girls develop a special attachment towards their fathers.

Latency Stage

This stage of the child spans from 7-11 years of age. Sexuality and its associated pleasures become dormant. The child explores individuals other than family members.

Phallic Stage

This final stage of the child development runs from 11 years to adulthood. Same gender hostility is suppressed, affection for the opposite sex increases.

Psychosexual Stages: Summary:



Oral: Age 0 to 2

Infant seeks oral gratification by sucking, biting, and babbling.



Anal: Age 2 to 3

Potty training helps toddlers balance their needs for anal gratification with society's demand to be clean and neat.



Phallic: Age 3 to 7

In early childhood an unconscious desire for the opposite-sex parent is controlled by identification with the same-sex parent.



Latency: Age 7 to 11

Sexual urges are repressed, and the child prefers same-sex companions.



Genital: Age 11 to adult

With puberty sexual urges reappear, and the adolescent learns about mature relationships.

Summary

We looked at the inherent nature of structural-organismic perspective and then discussed Freud's theory of psychosexual development.

THEORIES OF DEVELOPMENT

Piagetian Theory & Vygotsky's Sociocultural View

LESSON 10

Overview

Piaget's theory of child development

Vygotsky's Sociocultural View and some criticisms he has to offer against Piagetian theory.

Information-processing theories

Critical Questions:

How does Piaget's theory explain child development? What is the major premise of his theory?

What does Vygotsky's view proclaim about child development?

How is information-processing theory different from structural-organismic theories?

Binet-Simon Scale:

1. Easy tasks included, following a lighted match visually, or shake hands etc.
2. Slightly harder tasks required naming body parts, repeat a series of 3 digits, define words like "house" or "fork" etc.
3. More difficult test items required drawings from memory or constructing sentences from words.
4. The hardest tasks included repeating back 7 random digits, rhyming words, and answering questions involving complexity etc.

5. Number of successfully completed tasks determined the child mental age. If an eight year-old child passed all the items, usually passed by 8 year-olds, but nothing beyond, then this child would have a chronological and mental age equaling eight years.

Intelligence Testing:

1. When Piaget worked for Binet, he was fascinated by the mistakes children made on the scale, and found that children of the same age-group made the same kind of mistakes. These mistakes were qualitatively different for one age-group of children than for another.
2. Piaget employed open-ended questions (clinical method) to study children.

Piaget's Conclusions:

1. Piaget realized that intelligence cannot be measured by conventional intelligence tests.
2. The individual and the environment are changing constantly, so *intelligence* works as a mode (trait) to optimize individual's response in its survival.
3. Intelligent act leads the individual to deal effectively with his environment, i.e., make him adapt to his environment.
4. Piaget concluded that intelligence could not be inherited, but was a dynamic trait changing all the time due to biological maturation and the individual's experience.
5. This understanding finally culminated in what Piaget called *genetic epistemology* (1950), or the study of development of knowledge.

General Summary:

A general understanding of Piaget's theory is based on the idea that humans pass through stages of physical and intellectual development, as our bodies grow so does our intellect.



Cognitive Structure:

A cognitive structure consists of schemas (or schemata), and results from both biological maturation and cumulative experience. Therefore, a cognitive structure develops and grows over an individual's life span.

Schema:

A schema is an element of the cognitive structure, engages in a class of actions. So, grasping, sucking, reaching etc., are all schemas because a variety of actions can be made by them.

Actions:

A schema can lead to an overt action (behavioral response) or a covert action (thinking). So when an individual grasps a book that is an overt action. However, when one thinks of grasping an object (a golf club) that is a covert action.

Assimilation & Accommodation:

1. The process of responding to the environment in accordance with one's cognitive structure is called *assimilation*. In assimilation existing cognitive structures are matched with physical environment.
2. *Accommodation* is a process that modifies cognitive structure or more simply a process of learning.

Stages of Development

Age	Stage	Description
Birth to 2 years	Sensorimotor Stage	Reflex base, circular behavior, Object permanence
2 to 7 years	Preoperational Stage	Understanding of symbols, egocentric
7 to 11 years	Concrete Operational Stage	More than one point of view, no abstract problems.
11 years and above	Formal Operational Stage	Abstract thinking, reason logically.

Sensorimotor Stage:

A stage between birth-2 years, and is marked by infant's inability to express a formal language.

The infant begins to learn about the world through sensory and motor interactions. At first the infant carries out simple reflex actions (grasping) followed by voluntary actions of grasping.

Between 1-4 months, the child practices *primary circular reactions*, actions that serve as stimuli and responses. For example, sucking her thumb feels good, so she sucks some more.

Preoperational Stage:

However, the child is quite egocentric during this stage, i.e., she sees things pretty much from her own point of view. Piaget and Inhelder (1948) used the “three mountain test” to show egocentrism.

Period of intuitive thought (4-7 years) is a part of this stage in which problem solving is without logical rules, and thus children are unable to correctly carry out conservation problems.

Concrete Operational Stage:

By slightly 7-years most children, have the ability to conserve volume, length, mass and number. A child knows that number of marbles is equal in the long and the short column.

This stage lasts from 7-11 years; the child not only uses symbols but can manipulate them logically. Thus uses logical operations to solve problems. But the problems need to be concrete in context.

By 7-8 years, children learn logical operations can be reversed. So a big ball of clay can be divided into small balls and put back together. The mass of big and small balls remain the same. Similarly, reversibility of operations can work at the thinking level: If the child can add $3 + 5 = 8$, she can also subtract to verify the relation $8 - 3 = 5$.

By 9-10 years at last children master area conservation and by 10-11 years, the child learns *class inclusion*. When shown 8 orange and 4 green candies, and asked whether there are more orange candies or more candies, the child in this stage will say “more candies”. This stage also marks *transitivity* as a cognitive property or ability to make logical inferences on the basis of separate relationships develops at this stage, e.g., if John is taller than Frank, and Frank is taller than Bob, then John is taller than Bob. Finally, at this stage sorting and seriation problems are correctly solved.

Formal Operations Stage:

This stage ranges from 11 to adulthood. Now the individual is able to solve abstract problems, moving on from concrete operation problems. At this stage the individual starts to *systematically search for solutions* like the pendulum and other problems. In short *hypothesis testing* begins at this stage; adolescents and adults quickly find answers to pendulum problem, like what factors would determine the speed of pendulum. Is it the length of string, weights, or the initial swing? With hypothesis testing they work it out and find that it is the string's length that determines the speed.

Lev Vygotsky:

A Russian psychologist who suggested cognitive development as an apprenticeship in which children advanced by interaction with others more mature. Vygotsky died young (37) and did not fully develop his theory beyond childhood.

Vygotsky defined *zone of proximal development* (ZPD) as the distance between child's actual *developmental levels* to solve a problem to *potential development* to solve the problem with peer assistance. And

suggested teachers should attempt to keep students in this zone in order to achieve maximum achievement.

Evaluations:

Contributions:

- Qualitative and developmental aspects of learning
- Description of developmental stages of attaining knowledge and learning

Criticism:

- Theory is based on clinical insights. Some experimental evidence fails these insights.
- Developmental stages may not generalize to non-western cultures.
- Vygotsky says that intellectual development may be speeded-up with assistance before stage reaches its maturation.

Information Processing Theory:

A completely different approach about child development comes from information-processing theory which considers human mind just like a computer.

Just as information processing systems use flow-charts etc to solve problems, human mind also processes problem and learns from experience in the same way.

To study, how humans solved problems and use information in a reiterative way. A 5-year old was given a problem to build a bridge with blocks across an imaginary river down below.

Information theory criticizes Freudian, Piagetian, and Vygotsky's stage theories and suggests that learning, problem solving, and other cognitive processes are similar at all ages in a human, present in lesser or greater extent.

Summary:

We looked at Piaget's theory of child development; and Vygotsky's Sociocultural View along with some strengths and weaknesses in both theories.

Finally we briefly discussed Information-processing theory.

LESSON 11

THEORIES OF DEVELOPMENT

Behaviorism, Cognitive, Social Learning Theory

Overview:

We will discuss behaviorism and its many flavors and tie them to child development.

What does cognitive perspective have to offer for child development?

What is social learning theory? How does this theory differ from conventional behaviorism to explain child development?

Critical Questions:

What does behaviorism has to offer for child development?

How do cognitive and social learning perspectives provide us with an understanding of child development?

Behaviorism:

The school of behaviorism has its roots heavily grounded in animal physiology; and its classical form was originated by Pavlov, and augmented by Watson.

This form of behaviorism relies on reflex behaviors and its modification.

Classical Conditioning:

Modification of reflexive behavior entails a learnt association that occurs between a specific and neutral stimulus called classical conditioning.

Generalization:

Generalization refers to increased capability of producing a CR by stimuli that are similar to the first CS that lead to conditioning. After conditioned fear of the white furry rat was inculcated in "Little Albert," he was afraid of other stimuli that resembled the white rat, like a furry rabbit, a dog, or a Santa Claus mask, etc., (Watson & Rayner, 1920).

Cortical Mosaic:

This excitatory and inhibitory activity in the brain forms a cortical mosaic. *Momentary cortical mosaic* determines how an organism will respond to its environment at a given moment. And a *dynamic cortical mosaic* is a stable mosaic in the midst of changing environment,

Summary:

Classical conditioning theory suggests that an association builds between CS and the US leading to UR/CR. CS elicits CR, and this connection is mediated through an empty organism.

Operant conditioning (Thorndike and Skinner) on the other hand proposes that a response like salivation (CR) prepares the organism to receive food (US).

Verbal Behavior:

Like any other behavior, language is also a behavior (verbal behavior) and consists of speaking, listening, writing and reading behaviors. These behaviors are governed by antecedent conditions (stimuli), and consequences (reinforcements). Skinner outline four of such verbal behaviors listed below in the table.

ABC of Verbal Behavior:

Type	Antecedent (A)	Behavior (B)	Consequence (C)
Mand	State of Deprivation or aversive stimulation	Verbal utterance	Reinforcer that reduces state of deprivation
Echoic	Verbal utterance from another individual	Repetition of what the speaker says	Conditioned reinforcement (praise) from the other person
Tact	Stimulus (usually object) in the environment	Verbal utterance naming or referring to the object	Conditioned reinforcement from the other person
Autoclitic	Verbal utterance (often a question) from another person	Verbal response (answer to a question)	Verbal feedback or reinforcement

Comparison of Behaviorism and Gestalt psychology:

Behaviorism	Gestalt
Elements of behavior	Mind or behavior must be studied in “wholes”, not as elements or parts.
Molecular behavior	Molar behavior or mind
Observation and Experimentation	Introspection, Observation and Experimentation
Approach: Behavioral	Approach: Cognitive

Law of Prägnanz:

Law of Prägnanz (Prägnanz mean “essence”) is a tendency that makes every psychological event simple, concise, symmetrical, harmonious and complete.

Not only was this law used as a guiding principle for studying perception, but also memory, learning, personality and psychotherapy.

Productive Thinking:

To generate productive thinking, students should arrange and rearrange the problem in many ways until the solution emerges based on understanding.

Wertheimer added that learning with logic or S-R associations would lead to limited learning. He came up with a number of examples to show that.

Finding Areas:

Students can be taught to find areas of rectangles by using algebraic formulas like length X height (L X H). However, when students are given non-standardized figures, such formulas lead to erroneous answers.

The correct way to learn to find areas is to understand the concept of unit area or a grid. Unit area can then be applied to any standard or nonstandard figure, to calculate its area.

Bobo Doll Study:



<http://www.youtube.com/watch?v=vdh7MngntnI>

Vicarious Reinforcement:

Bandura showed in his bobo doll that children imitate model's behavior.

1. Behavioral learning has a cognitive explanation.
2. Learning did not require reinforcement.
3. Reinforcement (incentive) directs learned activity (performance).

Summary:

We looked at Behaviorism as a basis to talk about learning in children. We did the same for cognitive perspective and finally discussed social learning theory

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LESSON 12

THEORIES OF DEVELOPMENT

Dynamic Systems Perspectives; Contextual Perspectives; Socio-cultural Theory;
Bronfenbrenner's Ecological Theory

Overview:

Brief prelude to socio-cultural perspective of child development

Bronfenbrenner's Ecological Theory and contextual perspective on child development
The Dynamic Systems View of child development.

Critical Questions:

How is context so important in child development? Looking at many socially oriented perspectives
How does Dynamic Systems View help us integrate all these perspectives?

Socio-Cultural Theory:

Socio-cultural theory (Vygotsky) is relatively a new theory in psychology that explains important contributions *society* makes to individual development.

According to Vygotsky, "Every function in the child's cultural development appears twice: first, on the social level (inter-psychological), and later, on the individual level (intra-psychological)."

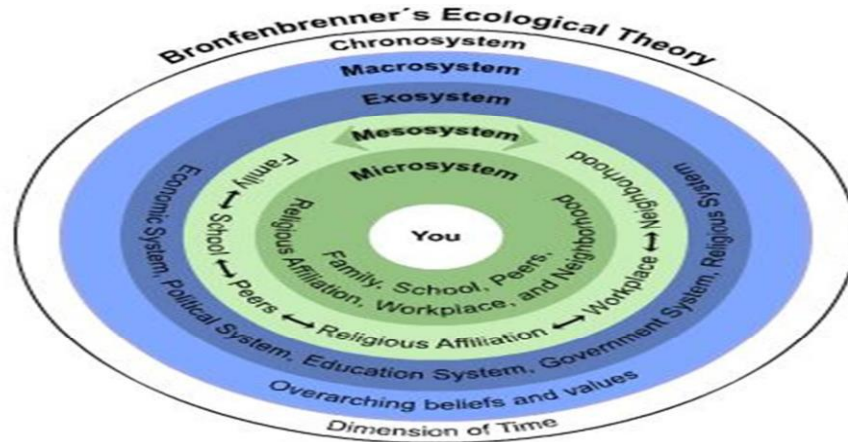
Principles:

According to Vygotsky:

- Development cannot be separated from its social context.
- Learning can lead to development.
- Children construct their knowledge.
- Language plays a vital role in mental development.

Ecological Systems Theory:

1. This theory centers on ecology of the child during development.
2. Bronfenbrenner proposes complex "layers" of environment, each affecting the child during development.
3. Changes in any layer of the environment ripples across other layers.
4. The interaction between child's maturing biology and these layers steers his development.
5. Children are both products and producers of their environment.



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The inner most layer of the environment constitutes the microsystem, which consist of immediate surroundings, including the person's family, peers, school, and neighborhood.

Next layer consists of mesosystem that holds relations between the different microsystems or connections between contexts. Some common examples are the connection between family experiences and school experiences.

The exosystem further removed from the individual, may directly or indirectly affect the individual's behavior. For example, what a wife or child experience at home may be influenced by the husband's experiences at work.

Macrosystem defines the culture in which individuals live. Socioeconomic status, poverty, and ethnicity make this culture.

Finally the last layer termed as chronosystem refers to the patterning of environmental events over the life of an individual and socio-historical events or circumstances.

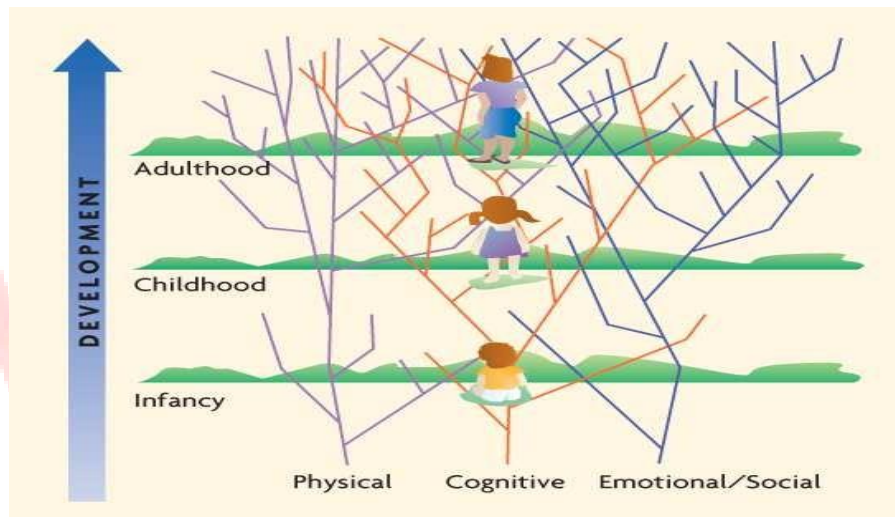
Contextual Perspective:

Both in Socio-cultural theory of Vygotsky, and Ecological System Theory of Bronfenbrenner propose context (society, environment) to play an important role in the developing child.

Although biology of the child is important, behavior is largely shaped by his context. This view heavily tilts on the nurture side of the nature-nurture equation.

Dynamic Systems View:

1. Dynamic System View takes an *integrated system* approach putting the child's brain, mind, body, physical and social worlds and investigates how the child masters new skills as these worlds change.
2. Dynamic System approach suggests that children do not change linearly in a line, but more like a web fibers branching out in many directions, each representing a evolving skill.
3. Dynamic System View has been inspired by other disciplines, like biology and physics, in addition draws on information-processing, contextual, sociocultural, ecological and evolutionary theories.



Summary:

We looked briefly at *Vygotsky's Socio-Cultural Perspective* of child development. We added more to this perspective by adding *Bronfenbrenner's Ecological System Theory* and *Contextual Perspective* on child development. Finally we discussed the *Dynamic Systems View* of child development.

LESSON 13

THEORIES OF DEVELOPMENT

Span Perspective; Ethological and Evolutionary Approach

Overview:

We will look at ethology and comparative psychology explaining child development. How does evolutionary theory help us understand child development?

Critical Questions:

Do you believe that traits like intelligence and personality are inherited or learned? Why do you believe in one or the other or both?

Ethology:

Ethology (or comparative psychology) is a field of inquiry that has its roots in biology and evolution.

Investigators like Lorenz and Tinbergen studied instinctive behaviors as model systems to understand adaptation. These instinctive behaviors became the basis to understand attachment behaviors in animals and children.

Instinctive Behaviors:

Instincts are complex pre-programmed genetically controlled behaviors also called Fixed Action Pattern (FAP), migration and nest building are but some its examples.



Instincts in Humans:

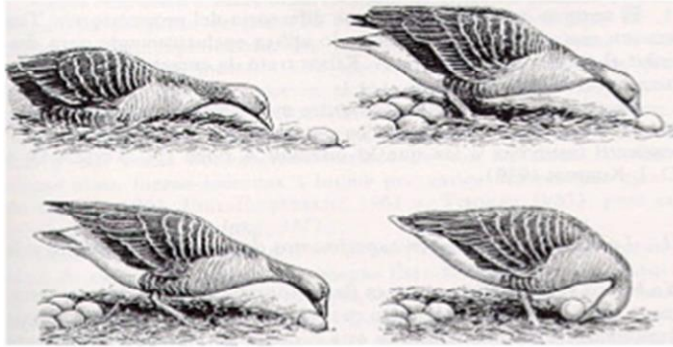


Humans also show many instinctive behaviors, e.g., smiling, crying, frowning, kissing, cuddling and aggressive behaviors. Included in these is the “Eyebrow-flash” during smiling (Eibl-Eibesfeldt & Hass, 1972; 1990).

Characteristics of Instincts:

1. Instincts are under innate genetic control. All geese roll the egg in the same way.
2. Instincts require little or no feedback. If hatchlings are lost, dominant “paternal instinct” leads cardinal to feed gapping minnows.

3. Instincts can be triggered by stimulating brain “trigger” cells.



4. Instincts require coordination of many muscles thus more complex than simple reflex actions.

Imprinting:

1. Imprinting is a process that modifies instinctive behaviors. Includes filial (following mother or a substitute) and sexual (courting member of another specie) imprinting.
2. Imprinting takes place during “sensitive period” of individual development.
3. Many behaviors are released (releasers) to make mothers and caregivers form close bonds of attachment.
4. These attachment bonds play a vital role in individual’s adult life.



Konrad Lorenz and
filial imprinting (1938)



Konrad Lorenz and
sexual imprinting (1938)

Sensitive Period:

So imprinting can take place during a window of opportunity early in life called the sensitive period. In humans too there is a sensitive period, an optimal time for certain capacities to emerge. The individual is especially responsive to environment. Developmental changes are hard to induce later. Boundaries less defined than during the sensitive or critical period

Harlow & Attachment:

Harry Harlow (1960) studied attachment in non-human primates. He, like Lorenz, suggested that attachment bonds developed not just because of feeding but other factors like bodily comfort and love.

Bowlby & Attachment:

Likewise Bowlby (1952) was influenced by Lorenz ideas of imprinting and went on to use ethological perspective in child-caregiver attachment.

“According to Bowlby, the infant's relationship to the parent begins as a set of innate signals that call the adult to the baby's side (Pendry, 1998)”.

Ainsworth & Attachment:

1. Ainsworth empirically used separation anxiety to study attachment between children and their caregivers, and found three kinds of attachments.
2. Infants were either securely attached to their caregivers; when separated they protested but greeted the caregiver on her return.
3. Or, infants were insecurely attached or were avoidant; for they did not display distress at separation.
4. The third kind of infants, were ambivalent or anxiously attached and showed great deal of anxiety at separation and anger and resistive behavior at the caregiver's return.

Evolutionary Developmental Psychology:

Expands upon ethology

Seeks to understand adaptive value of human competencies

Studies cognitive, emotional and social competencies and change with age

Summary:

We looked at ethology and comparative psychology explaining child development. We also discussed Harlow, Bowlby, and Ainsworth theories. Lastly we touched upon how evolutionary theory helped us understand child development.

LESSON 14

THEORIES OF DEVELOPMENT

Stages of Prenatal Development

Overview

We will look at the process of germination and fetal growth.

We will also discuss problems and diseases that take place during prenatal development.

Critical Questions

How does germination take place?

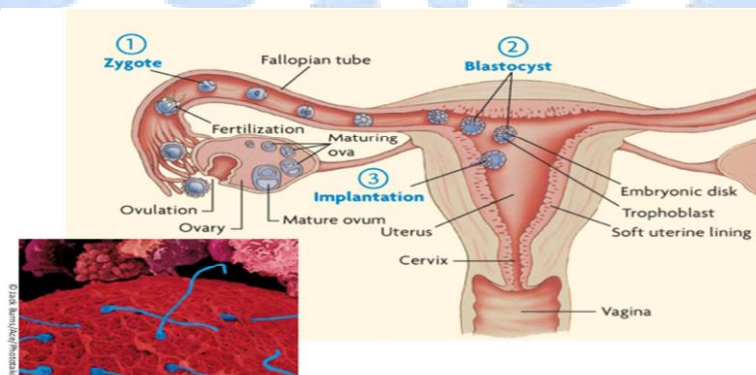
How does the embryo-fetus develop? What are some of the things that we need understand about prenatal development?

What are some of the risks in prenatal development?

Prenatal Development:

Period	Duration	Key events
Zygote	0-2 weeks	Fertilization, implantation, start of placenta
Embryo	3-8 weeks	Arms, legs, face, organs muscles start developing. Heart begins beating
Fetus	9-30 weeks	Growth and finishing

Germination:



Conception & Differentiation:

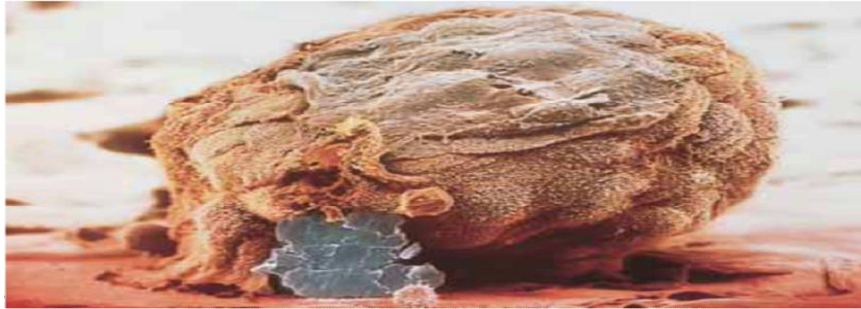
Ovulation: Release of an egg (ovum) from the female ovary.

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Conception: The process of fertilization where a sperm cell combines with an egg cell to create a new organism (zygote).

Differentiation (embryonic induction): Process that occurs during cell division in which each new cell is committed to becoming a particular structure.

Zygote: 7-9 days old



Food & Shelter

Chorion: Protective membrane that surrounds the amnion.

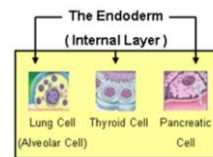
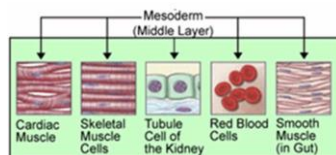
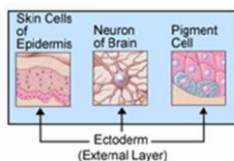
Placenta: Permits food and oxygen to reach developing organism, and removes waste products to be carried away.

Umbilical Cord: Placenta grows into umbilical cord. Delivers blood to developing organism. Removes waste from developing organism.

Embryonic Growth:

1. Last Half of the First Month

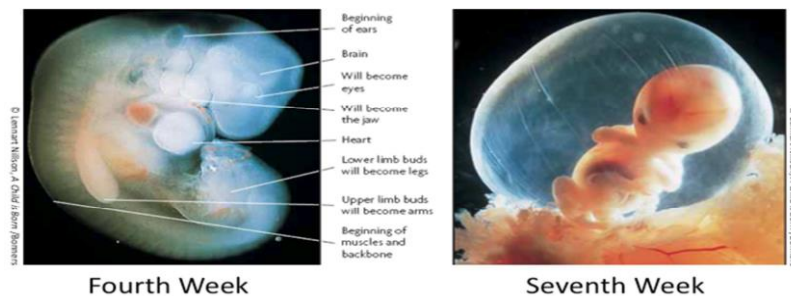
- Ectoderm
- Mesoderm
- Endoderm
- Neurons



2. Second Month

- Rapid growth: eyes, ears, nose, jaws, neck
- Buds become arms, legs, fingers, and toes
- Responds to touch

Embryo:



Fetal Period:

Third Month

Lungs begin to expand and contract

Sex is evident on ultrasound

The Second Trimester

Mother can feel movements

Vernix, lanugo develops

Brain weight increases tenfold from 20th week to birth.

The Third Trimester

Age of viability: 22-26 weeks

Cerebral cortex enlarges

Fetus spends more time awake

Fetus at 11 weeks:

The fetus at 11 weeks can kick, bend its arms, open and close its hands and mouth and suck its thumb.



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Second & Third Trimester:



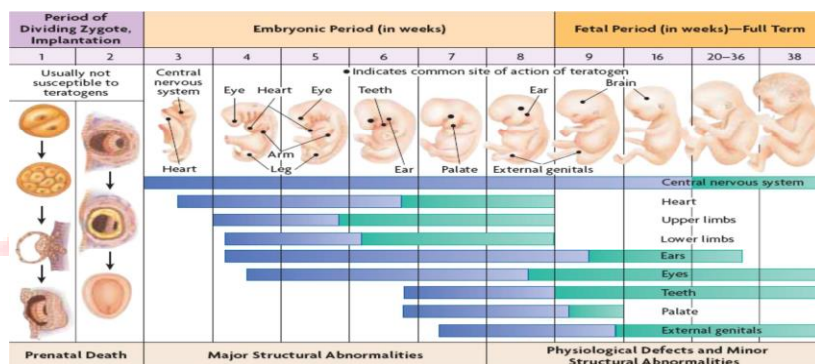
Twenty-second week



Thirty-sixth week

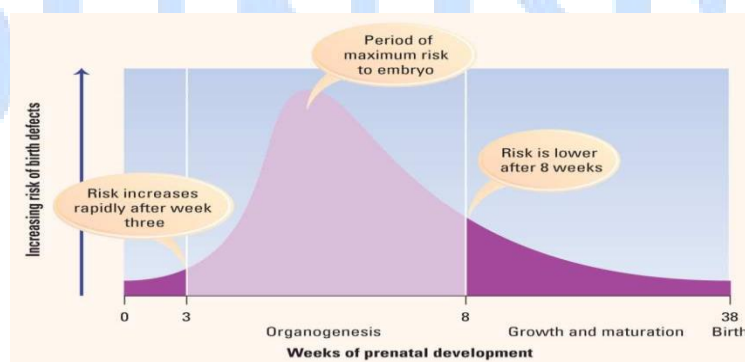
Sensitive Periods:

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Sensitive Period: Risks:

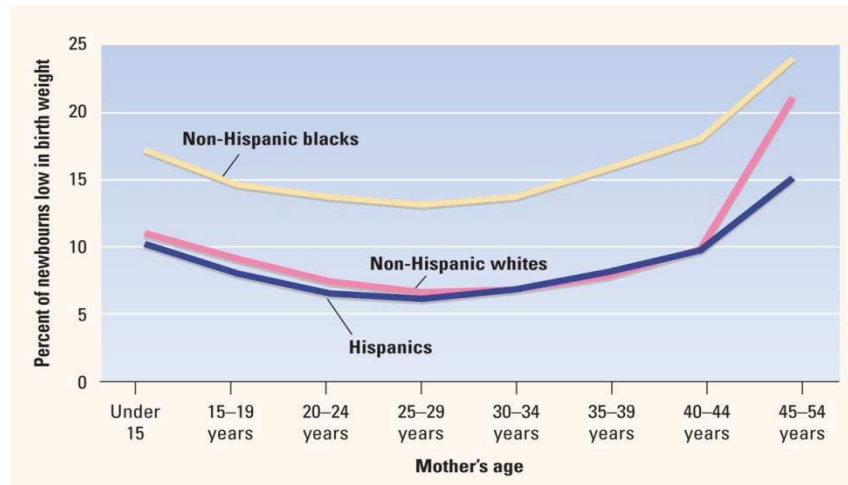
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Prenatal Development & Later Health

1. Low Birth Weight and greater chance of
 - Heart disease
 - Stroke
 - Diabetes
2. High Birth Weight and greater chance of
 - breast cancer

Low Birth Weight: Ethnicity:



Teratogens:

Teratogens are any kind of drug or substance that interferes with the development of fetus, usually causing birth defects.

Teratogens: Risks to the Baby:

Premature: Babies born earlier or smaller than average.

Preterm Birth: Births that occur before 37 weeks of gestation.

Low Birth Weight: Weight less than 5 ½ pounds at birth (2 pounds lower than average).

Alcohol:

Alcohol use during pregnancy is catastrophic.

Fetal Alcohol Syndrome (FAS): A syndrome of birth defects caused by prenatal exposure to alcohol. Fetal Alcohol Effects (FAE): Lowered IQ, hyperactivity, growth deficiencies, and physical malformations.

Cocaine & Smoking:

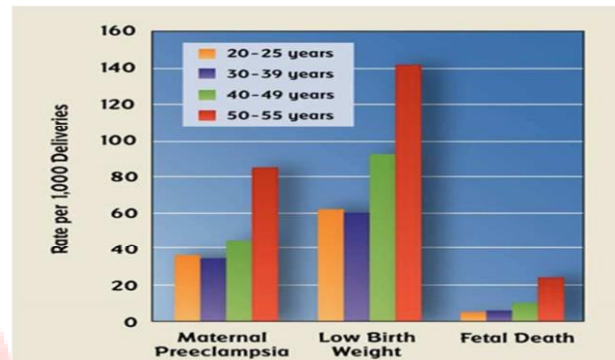
Cocaine: Fetal exposure to cocaine retards growth can cause preterm birth, and malformations in the brain, intestines, and genital-urinary tract.

Cigarette Smoking: Harmful chemicals, including nicotine, carbon monoxide, and cyanide. These chemicals damage the placenta; reduce the blood supply, oxygen, and nutrients available to the fetus.

Viral & Infectious Diseases

If a woman is carrying an infectious disease, like herpes, syphilis or HIV-AIDS while pregnant, it can endanger the fetus.

Maternal Age & Birth Complications



Healthy Prenatal Development:

- Exercise
- Nutrition
- Lower Emotional Stress
- Younger Age
- Spaced Previous Births

Summary:

We discussed the process of germination and fetal growth. We also discussed the problems and diseases that take place during prenatal development.

LESSON 15

PRENATAL DEVELOPMENT AND BIRTH

Birth and the Beginnings of Life

Overview:

In this lecture we will look at child's birth and beginning of his or her life.

We will also look at what are some of the early events that take place in the neonate's life. These will include birthing complications, vulnerabilities, and early human associations.

Critical Questions:

Why do you think the process of birthing is an important event for the mother and her child? How do we assess the neonate?

Natural or Prepared Childbirth:

1. Classes
2. Relaxation & Breathing Techniques
3. Labor Coach and Birth Attendants
 - Certified nurse-midwives
 - Doula (social support)

Natural or Prepared Childbirth

Most child deliveries in Pakistan are assisted through midwives or medical staff. Extended families provide support for prepared childbirth.

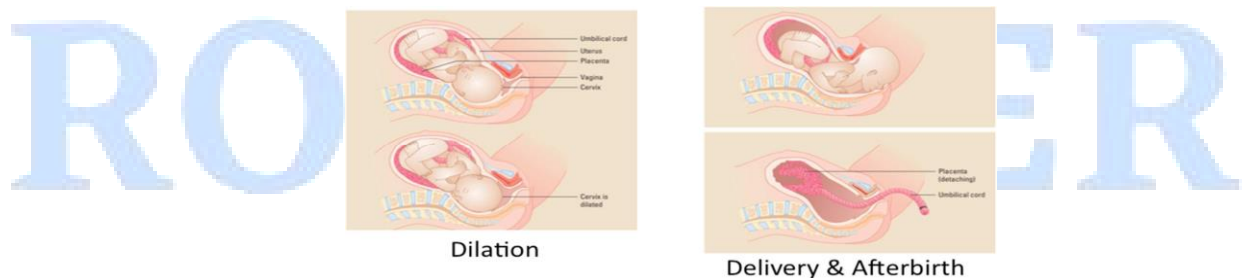
The Stages of Birth:

Stage 1 Dilation: The gradual opening of the cervix caused by labor contractions.

Stage 2 Delivery: The baby actually moves through the birth canal.

Stage 3 Afterbirth: The placenta and other membranes are delivered through the birth canal.

The Stages of Birth:



Labor and Delivery Medications:

Some form of medication is used in more than 80% of North American births.

1. Analgesics
2. Anesthetics
3. Epidural analgesia

Birth Complications:

1. Oxygen Deprivation (Anoxia)
2. Breech Position: bottom down position.
3. Placenta Abruptio: the separation of placenta from uterus wall before baby is delivered.

4. Rh Factor Incompatibility.
5. Fetal Distress

Long-Term Effects of Anoxia:

After initial brain injury, another phase of cell death can occur several hours later.

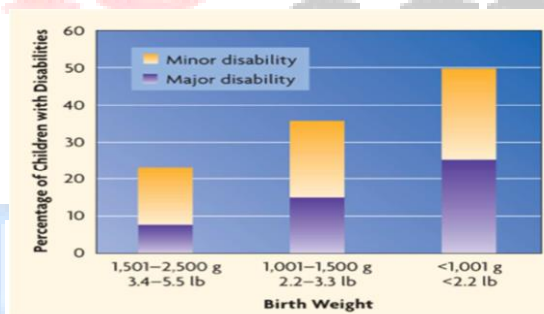
To prevent secondary damage, researchers are using head cooling devices and pre-cooled water blankets.

The greater the oxygen deprivation, the poorer children's cognitive and language skills in middle childhood.

Preterm & Small-for-Date Babies:

- Preterm: Born weeks before their due date. May be appropriate weight for length of pregnancy.
- Small-for-Date: May be born at due date or preterm. Below expected weight for length of pregnancy.

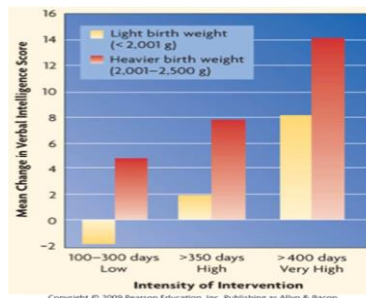
Low Birth Weight & Disabilities:



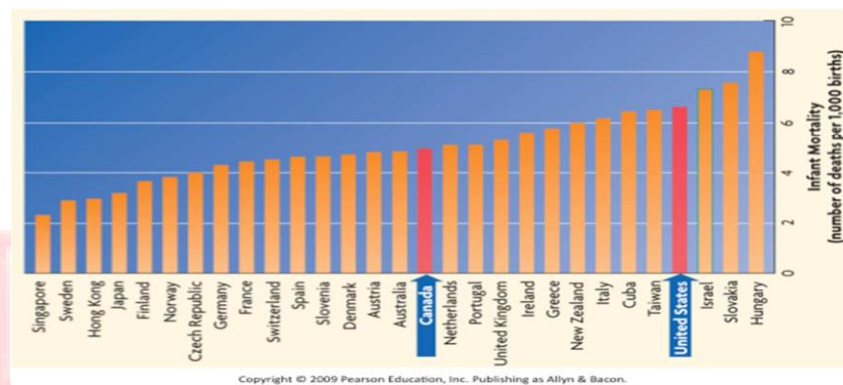
Interventions for Preterm Infants:

1. Isolette Respirator
Feeding tube
Intravenous medication
2. Special Infant Stimulation
"Kangaroo Care"
3. Parent Training

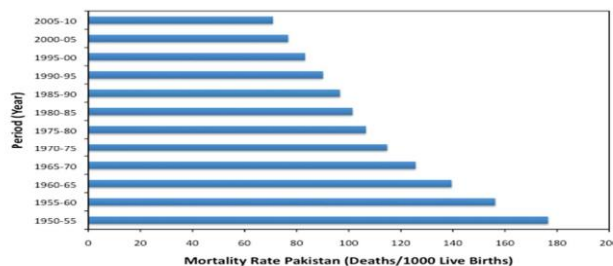
Intervention Intensity:



Infant Mortality: World:



Infant Mortality: Pakistan:



The New Born!

APGAR Test: A brief assessment of the newborn conducted at 1 and 5 minutes after birth; used to identify newborns that are at risk and need medical attention.

Becoming a Family: Parenthood

It usually takes about 6 months to adjust to a new baby (although time may vary).

The transition to parenthood often brings to the forefront any issues that a new parent has with his or her own parents.

The transition is easier if the new parents are realistic in their expectations.

Becoming a Family: Siblings

An older sibling may have a difficult time with a new baby.

- Increases in whininess
- Sleeping difficulties
- Withdrawal
- Clinginess
- Aggressive behaviors

Summary

In this lecture we saw child's birth and beginning of his or her life.

We also looked at what are some of the early events that took place in the neonate's life.

LESSON 16

PRENATAL DEVELOPMENT AND BIRTH

Socio-cultural Perspective

Overview:

In this topic we will look at newborns with special interest in interpersonal processes that are formed and maintained with parents, caregivers, and siblings.

We will also look at variety of research that addresses physiological, psychological and social aspects of these bonds during early development.

And introduce the field of prenatal and perinatal psychology.

Critical Questions:

What kind of relations are formed and maintained between the newborn and his family members? How good is our understanding of these relationships?

Prenatal Bonding:

1. The mother and the prenatal baby form an early bond. The bond is highly emotional, and generally tend to affect the unborn baby at an emotional and physiological level.
2. A variety of emotional experiences directly affect the prenatal baby physiologically via the placenta.
3. Not long ago doctors thought it is not necessary to bond with the baby.
4. But today many researchers and experts think that human voices, communication, and music is required in establishing a strong bond between parents and prenatal baby.

Prenatal Psychology:

Is an interdisciplinary field that explores the psychological and psychophysiological effects and implications of the earliest experiences of the individual, before birth ("prenatal"), as well as during and immediately after childbirth ("perinatal").

It also looks at the health and learning ability of the individual and on their relationships.

Prenatal Psychology: History

Otto Rank became convinced that birthing trauma led to neurotic behavior. Wrote "The Trauma of Birth (1932)" and developed psychoanalysis around it.

Birth Without Violence (1975), by Frederick Leboyer became the basis of "Leboyer and Odent baths".

Prenatal Stimulation:

1. Prenatal stimulation enhances visual, auditory, and motor development.
2. Improves learning and growth. Babies tend to have higher intelligence and creativity.
3. Provides babies with early bond of trust and love.
4. Stimulated babies rate high on APGAR scores and have higher IQs.
5. Good head and general movement control. The infants show more alertness, confidence and peacefulness.

Brain Stimulation:

When the brain of a neonate is stimulated new neuronal connections are established.

This determines intelligence, social, and emotional skills of the child. And has profound effects on human personality and aptitude.

Brain Development:

In fifth week after conception, the first synapses form (spinal cord). By the sixth week, first fetal movements are regulated by them.

Other movements limbs (8 weeks) and fingers (10 weeks), as well as some surprisingly coordinated actions (hiccupping, stretching, yawning, sucking, swallowing, grasping, and thumb-sucking) around 12 weeks.

Brain Development:

Most women sense the first fetal movements around 18 weeks of pregnancy.

The second trimester marks the onset of other critical reflexes like diaphragm and chest muscles and coordinated sucking and swallowing reflexes controlled by the brainstem.

Heart rate, breathing, and blood pressure then follow culminating in conscious experiences at the end of the third trimester

Other Family Members:

Siblings and grandparents are the next set of individuals that form close bonds with the prenatal and perinatal baby.

Younger siblings are more effective and form close bonds than older siblings. But the picture is complicated.

Likewise grandmothers (maternal) form closer bonds with the prenatal baby than other grandparents

Siblings:

An older sibling may have a difficult time with a new baby.

Increases in whininess

Sleeping difficulties

Withdrawal

Clinginess

Aggressive behaviors

Sibling Rivalry:

Sibling rivalry is common among many cultures. Younger siblings (same gender) form competitive relations with the prenatal and postnatal baby.

Sibling rivalry is generally affected by parental attention, birth order and personality.

Freud suggested this rivalry was due to oedipal forces, sibling competed for the same parent. Other biological and evolutionary causes include competition for resources.

Summary:

In this topic we looked at newborns with special interest in interpersonal processes that are formed and maintained with parents, caregivers, and siblings.

We also looked at variety of research that addresses physiological, psychological and social aspects of such bonds during early development and the field of prenatal and perinatal psychology.

LESSON 17

INFANCY: SENSATION, PERCEPTION AND LEARNING

Physical Development of the Newborn

Overview

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We will discuss birth of the child in this section, with special emphasis on premature birth and infant mortality due to a number of diseases that take place in infants.

The lecture will also look at brain and body development and early learning. This brain-body development will include sensory and motor development.

Critical Questions

What kinds of infants are at risk when born? What do we understand by prematurity of birth and Infant mortality?

How does the body and the brain of an infant grows?

How do sensory and motor capabilities develop in the neonate?

What is Prematurity?

Normal gestational period of fetus is 280 day or 40 weeks (Normal range 38-42 weeks). A birth of a baby that takes place before 37 weeks of gestation is considered preterm and is generally premature.

Period	Second trimester										Third trimester																		
Week	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46
Term					Preterm																Normal		Post-Mature						

On or before 24 weeks the chances of survival are 50% for a fetus, because many organs of the body are not mature. A surviving neonate suffers from many health risks.

What is Prematurity?

Most organs in preterm births that occur between 34-37 weeks have optimal level of functioning, except the lungs. Lungs mature in the womb, and if not mature, the baby needs to be raised in a ventilator for lungs to mature.



It is generally said that preterm babies are premature and term babies are mature. In other words prematurity is inversely related to gestational age.

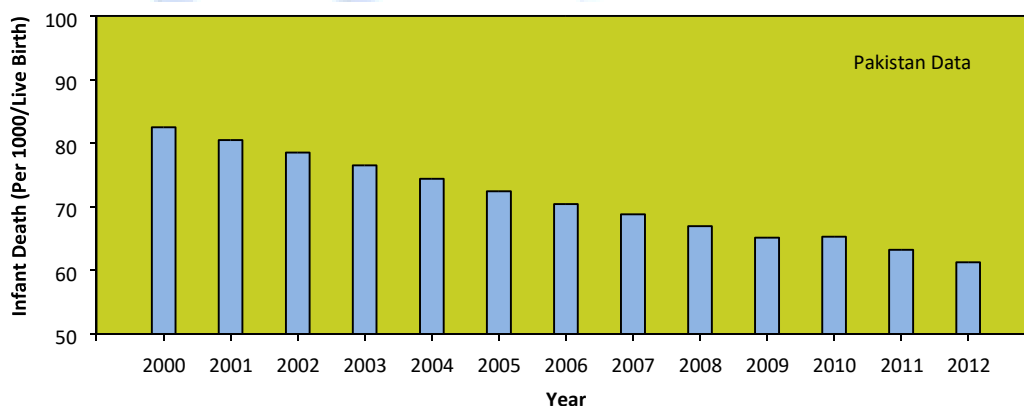
Prematurity and Health Problems

1. **Neurological Problems.** Many neurological problems including apnea, retinopathy, cerebral palsy and intra-ventricular and brain hemorrhage, white matter abnormalities causing language dysfunction are common in preterm babies. Preterm children are at a greater risk for having poor connectivity among many brain regions leading to a variety of learning disabilities.
2. **Cardiovascular and Respiratory Problems.** In preterm babies cardiac valve malfunction and chronic lung disease is common.
3. **Gastrointestinal and Metabolic Issues.** These issues include hypoglycemia, feeding difficulties, rickets of prematurity, hypocalcaemia, inguinal hernia, and necrotizing enterocolitis.
4. **Hematologic Complications.** Include anemia of prematurity, thrombocytopenia, and hyperbilirubinemia (jaundice) that can lead to kernicterus.
5. **Low Birth Weight.** Weight less than 3 ½ pounds (below the 10th percentile) at birth lead to greater potential for health risks.

Infant Mortality

Preterm babies are at a greater risk of dying between birth and 1 year of age. Infant mortality is related to:

- poor or absent prenatal care,
- teenage pregnancy,
- poor nutrition,
- risky health behaviors during pregnancy (drug abuse),
- higher rates of prematurity and
- low-birth weight births.



Prenatal Care

Access to and appropriate use of good-quality prenatal care (hospitals, neonatal nursing homes) results in healthier babies. Another important component of prenatal care is education. Education can increase the mother's knowledge about prenatal care.

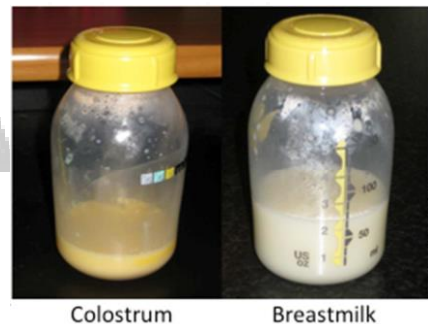


Feeding and Nutrition

Nearly all health officials agree that breast milk provides the best form of nutrition for most infants.

Colostrum – A thick, yellowish substance in breast milk containing important antibodies. Babies who are breastfed are at a lower risk for a variety of conditions.

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Thinking Critically

Think about any families you know that have young babies.

Are they feeding breast milk to their babies, or are they using infant formula?

What are the main factors that you think led to their choice?

Physical Growth

At birth, the average newborn weighs 7 ½ pounds.

Infants double their weight by the time they are 5 months.

The Nervous System

The brain and the spinal cord together form the central nervous system.

The nervous system gives rise to all of our thoughts, emotions, and behaviors.

The most complicated organ in the body, the brain is one of the first structures to form.

Neurons – are specialized cells that process information and allow communication in the nervous system.

The Nervous System

Major Structures in the Brain and Spinal Cord

The Nervous System

Synaptogenesis. One form of neuron maturation in which dendrites and axons branch out to form connections with neurons.

Myelination. A form of neuron maturation in which the fatty insulation grows around the axons.

Apoptosis. Programmed cell death. Process by which many neurons die during periods of migration and heavy synaptogenesis.

Sleep Patterns

During the first month, newborns spend two-thirds of their time sleeping.

This pattern slowly changes to an adult pattern of sleep (8 hours) between 5-9 years of age.

Co-sleeping – In many cultures infants and young children sleep with one or both parents.

Sleeping with infants and children give caregivers rest and strength so that they can continue with their care when the infant or child wakes up.

Sensory Capabilities

Seeing (Vision)

How clear is their vision?

Visual acuity – the ability to see fine detail.

Researchers estimate that visual acuity in newborns is somewhere between 20/150 and 20/600, and reach 20/20 by 6 to 12 months.

Can they see different color?

Color vision is relatively mature by 6 months.

How deep is that drop? Early depth perception.

Depth perception is available by the time infants learn to crawl.

Hearing (Audition)

Even before birth, fetuses react to loud noises.

By 6 months of age infants respond to a broad range of sounds including rattles, voices, songs, and many other environmental noises.

Smell and Taste (Olfaction and Gustation)

Immediately after birth, their facial expressions show that newborns react to certain odors in a manner similar to adults.

They express pleasant facial expressions to sweet and fragrant smells and annoyance to putrid and burnt smells. Similarly, infants show taste preferences immediately after birth, even before their first feedings. Like sweet, and dislike bitter tastants.

Motor Capabilities

Reflexes

Reflexes are involuntary movements that are elicited by environmental stimuli.

Rooting reflex. If you touch a newborn's cheek, the infant's head will turn in the direction of the touch.

Sucking Reflex. If anything touches an infant's lips, the infant automatically begins to suck.

Grasping Reflex. When an object touches an infant's palm; the baby's fingers will automatically wrap around the object and grip.

Voluntary Movements

Gross Motor Development. Process of coordinating movements with the large muscles in the body. Fine Motor Development. Process of coordinating intricate movements with smaller muscles. Proximodistal

Palmar

grasp

Pincer

grasp

Tripod grasp

Stepping reflex

Early Experiences

Parenting practices differ across cultures, infants in some cultures receive more vigorous physical stimulation than do babies in other cultures.

Toilet Training

Toilet training develops from an interaction of physical maturity, cognitive understanding, cues and feedback from the environment, and motivation.

Most toddlers gain voluntary control and coordination over the muscles that control their bladder and bowel movements by the time they are 18 to 24 months old.

Summary

We discussed birth of the child in this section, with special emphasis on premature birth and infant mortality due to a number of diseases that take place in infants. We also looked at brain and body development and early learning. This brain-body development includes sensory, motor and other experiential developments.

LESSON 18

INFANCY: SENSATION, PERCEPTION AND LEARNING

Sensory and Perceptual Capacities

Overview:

We will discuss early sensory and perceptual processes, especially the sense of touch and taste.

We will also include in our discussion intermodal processing, and explanation of intermodal processing.

Critical Questions:

What are some of the stages through which the individual passes to early in life?

What are some considerations we need to keep in mind with reference to normal and abnormal development in early life?

Infants' Sense of Touch:

- Newborn's reflex responses to touch on mouth, palms, soles, genitals
- Later, exploratory mouthing
- Pleasurable touch releases endorphins

Sensitive to pain

- Pain can affect later behavior.
- Relieve pain with anesthetics, sugar, gentle holding.

Taste and Smell:

- Infant prefers sweet tastes at birth and quickly learns to like new tastes.
- It has preferences of odors from birth and is affected by mother's diet during pregnancy
- Infant can also locate odors and identify mother by smell from birth.

Sense of Hearing

- Infant can hear a wide variety of sounds at birth
- Infant prefers complex sounds to pure tones.

- It learns sound patterns within days.
- It is sensitive to voices and biologically prepared to learn language

Developments in Hearing:

Four to Seven months	Sense of musical phrasing
Six to eight months	“Screen out” sounds from non-native languages. Recognize familiar words, natural phrasing in native language
Eight to nine months	Detect syllabus that often occur together in the same world.

Sense of Vision:

- Least developed of senses at birth
- Unable to see long distances, focus clearly
- Scan environment and try to track interesting objects
- Color vision improves in first two months.

Improvements in Vision:

- Brain development helps infants reach adult levels of vision.
- At 2 months, focus and color vision develops.
- At 6 months, acuity, scanning & tracking.
- At 6–7 months, depth perception.

Stages in Depth Perception:

Birth to one month	Sensitivity to kinetic cues
Two to three months	Sensitivity to binocular cues
Five to twelve months	Sensitivity to pictorial cues, wariness of heights

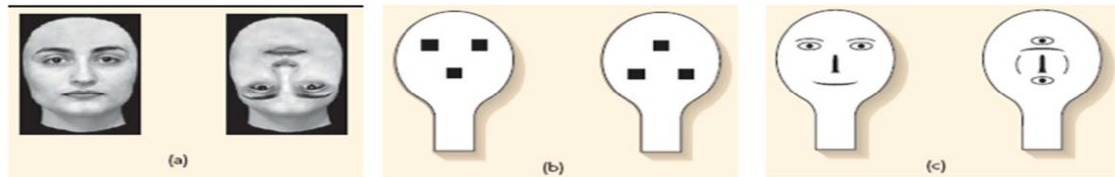
Stages in Pattern Perception:

Three weeks	Poor contrast sensitivity and prefers large simple patterns
Two months	Can detect detail in complex patterns, scans internal features of patterns
Four months	Can detect patterns even if boundaries are not really present
Twelve months	Can detect objects even if two-thirds of drawing is missing

Face Perception:

Newborns prefer simple face-like objects (left in panel b) over the one on the right. This ability goes away in about 6-months.

Infants prefer regular face to a scrambled face (panel c).



Size Perception:

Infants by 1 week of age have the ability to perceive differences in size.

Object Unity:

Infants 2 months old can perceive object unity.

Stages in Intermodal Perception:

Birth	Largely amodal sensory perception
Three to four months	Prefer “matching” sights and sounds
Five to six months	Reach for object in the dark, coordinating sight and touch

Differentiation Theory:

Differentiation theory proposes that infants search for *invariant* features of the environment.

And, note *stable relationships* between features, like visual patterns, intermodal relationships, etc. Gradually detect finer and finer features. *Differentiation!*

Affordances:

Gibson (1966) proposed that individuals discovered invariant features of the environment by acting on it. The environment “affords” such features and makes actions future-oriented.

Summary:

We discussed sensory and perceptual processes, especially with reference to touch, smell and taste, hearing and vision. We also discussed amodal and Intermodal processing and its explanation.

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LESSON 19

BRAIN, BODY, MOTOR SKILLS AND SEXUAL MATURATION

Nervous System and Brain Development in Infancy

Overview

We will look at major milestones the development of the brain in the neonatal baby.

We will also discuss major functions of the brain as they emerge when the brain develops.

Critical Questions

Why should we study the brain? Why is this organ so important? What should we understand about the brain as it develops?

Nervous System

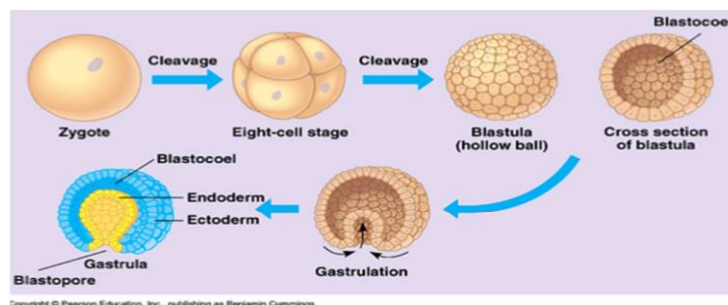
1. Nervous system is the most complicated piece of machinery in the universe, more complicated than a supercomputer.
2. Adult nervous system contains about a trillion of neurons, with many more neuronal connections.
3. The nervous system is divided into two parts, the central nervous system involved with higher brain functions and peripheral nervous system engaged in carrying out lower brain functions, like reflex actions.
4. Nervous system consists of neurons and glial cells. Neurons communicate within nervous system linking the outside and inside worlds.
5. Glial cells provide nutrition and support to neurons.

Neuronal Communication:

- Neurons communicate with one another using electrochemical messaging.
- They communicate with sensory organs and muscles and other organs in the body, to make the individual interact with the environment.
- Communication in neurons takes place in the form of action potentials and synaptic potentials.

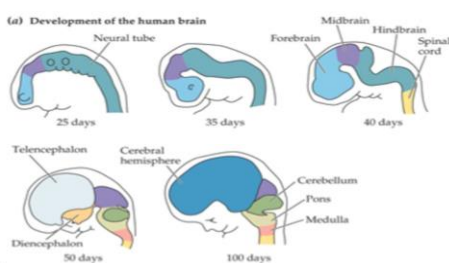
Embryonic Induction:

Embryonic induction leads to development of three layers in the embryo. Ectoderm results in forming the nervous system.



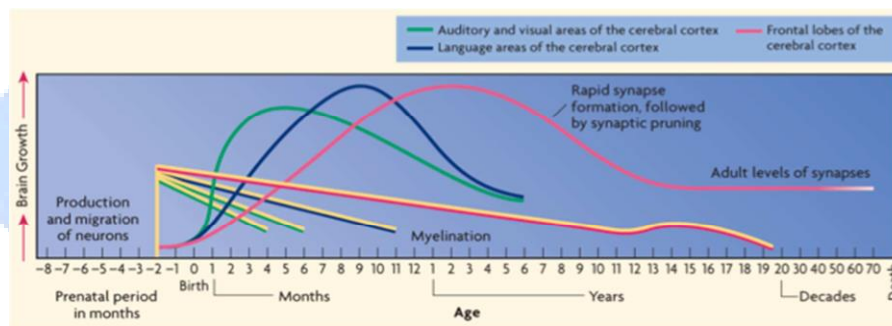
Brain Development:

The brain begins as a tube (neural) that balloons with development into separate compartments, forming different parts of the nervous system.

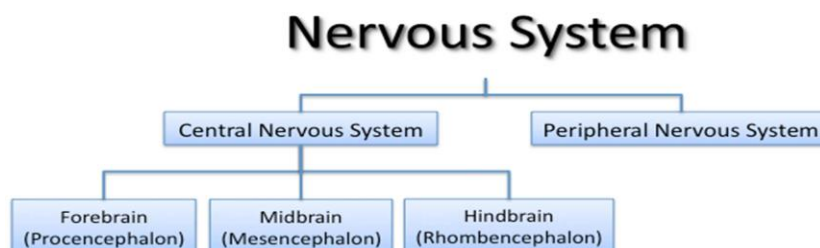


Brain Development: Milestones

The brain begins as a tube (neural) that balloons with development into separate compartments, forming different parts of the nervous system.



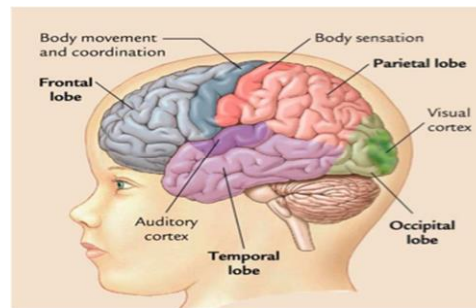
Nervous System:



Cerebral Cortex:

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Each hemisphere of the cortex is divided into four lobes. Each lobe is responsible for specific sensory and motor functions.



Lateralization of Cerebral Cortex:

Left Hemisphere	Right hemisphere
Sensory information and control of right side of body	Sensory information and control of left side of body
Verbal abilities	Spatial abilities
Positive emotion	Negative emotion
Sequential, analytical processing	Holistic, integrative processing

Brain Plasticity:

- In infants and young children, parts of brain are not yet specialized.
- Recover better from brain injury.
- Language recovers better than spatial skills.
- Still have some problems with complex mental skills.
- Older children, even adults, have some plasticity in the brain.

Handedness:

Reflects dominant cerebral hemisphere

- Right-handed (90%) - left hemisphere
- Left-handed (10%) - both hemispheres

May be genetic basis, but affected by experience

- Position in uterus
- Practice

Few left-handers show developmental problems.

- Left hemisphere damage may link left-handedness & some mental problems

Brain Development: Sensitive Periods:

Stimulation is vital when brain is growing rapidly.

1. Experience-Expectant Growth: (Ordinary experiences “expected” by brain to grow normally).
2. Experience-Dependent Growth: (Additional growth as a result of specific learning experiences).

Summary:

We discussed major milestones the development of the brain in the neonatal baby.

We also talked about major functions of the brain as they emerge when the brain develops.

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LESSON 20

BRAIN, BODY, MOTOR SKILLS AND SEXUAL MATURATION

Nutrition, Physical Growth,
Sexual Maturation & Sleep Patterns

Overview:

We will look at major nutritional needs of the child and associated physical growth.

We will also take a look at sexual maturation.

Finally we will take a look at sleep patterns of the growing individual.

Critical Questions:

What is balanced nutrition for a growing child? How do cultures regulate this balanced nutrition for the growing child?

How does an individual become sexually mature? What are gender differences in sexual development? How sleep patterns change in an individual? How has modern life affected sleep regulation?

Allometric Development:

Body proportions change over development. An infant 15 months old has a bigger head size compared to the body than an adult.

Growth Curve: Gender Differences

Between ages 6-8 years boys grow faster than girls. This reverses between 10-14 years.

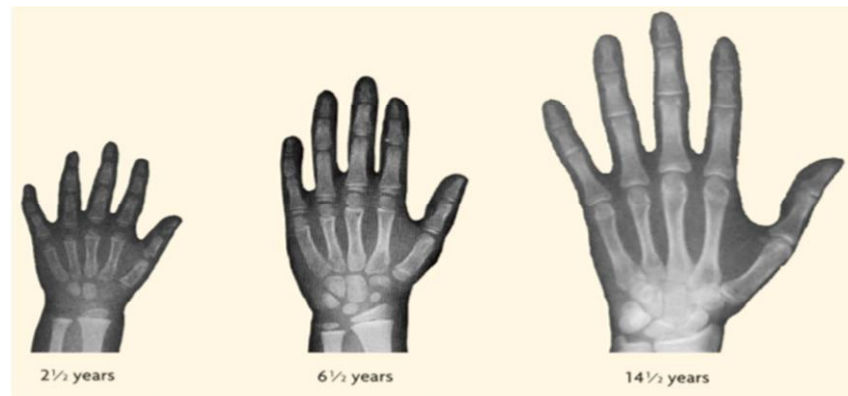
Finally boys after 14, attain higher greater height than girls.

Girls display growth spurt around 11 years and boys just before 14.

Fat-Muscle Makeup:

Period	Fat	Muscle
Birth/infancy	Fat peaks at 9 months in boys and girls. Girls have more fat than boys	Muscles are not added slowly
Childhood	From around age 8, girls add more fat on arms, legs, trunk and hips than boys	Muscles are added slowly
Adolescence	Girls continue to add fat but boys lose it	Boys add much more than girls

Skeletal Maturity:

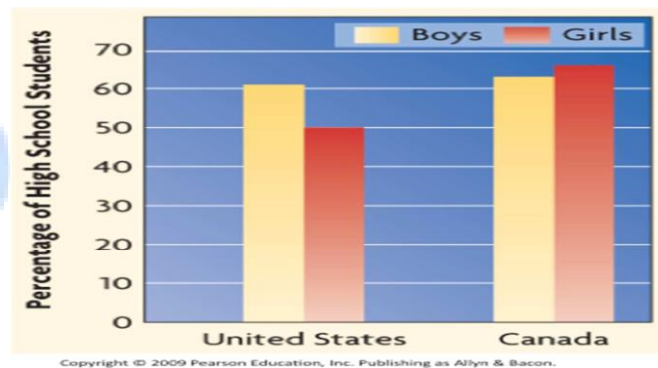


Motor Skills: Gender Differences:

1. Physical: During childhood, there are small differences across gender. During adolescence, boys develop more strength, speed, and endurance than girls.
2. Social: There is more athletic pressure on boys and may lead to more practice.

Sports Participation: Culture

Participation in sports differs across gender. In the US more boys than girls participate in sports. The opposite is true in Canada.



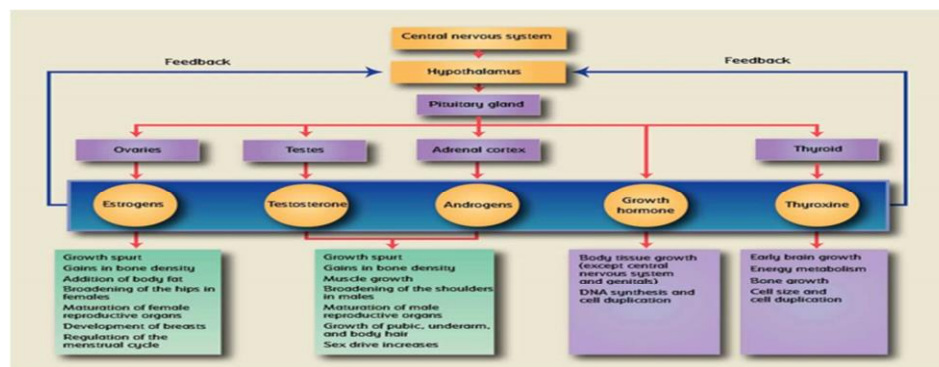
Benefits of Team Sports:

1. Regular physical activity.
2. Greater social competence, self-esteem.
3. Parents and coaches must emphasize
 - Effort
 - Improvement
 - Teamwork
 - *Not* competition

Hypothalamus & Pituitary Gland:

At the base of the brain there are two glands that regulate hormones. Among others these hormones regulate growth.

Hormonal Influences on Growth:



Hormonal Changes in Puberty:

1. Growth hormone & thyroxine increase around ages 8 to 9 years of age.
2. Estrogens (More in girls along with adrenal estrogens)
3. Androgens (More in boys, along with testosterone).

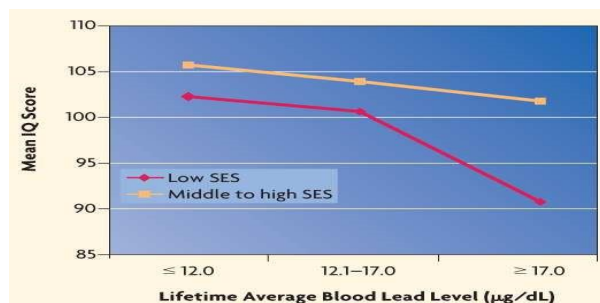
Worldwide Variations in Body Size:

1. Shortest children (South America, Asia, Pacific Islands, parts of Africa)
2. Tallest children (North & Central Europe, Australia, Canada, U.S.)
3. Also ethnic variations in *growth rate*.

Factors That Affect Physical Growth

1. Heredity.
2. Nutrition.
3. Infectious Diseases
4. Emotional Well-Being

Lead Exposure and Growth:



Benefits of Breastfeeding

- Correct fat-protein balance
- Nutritionally complete
- More digestible
- Better growth
- Disease protection
- Better jaw and tooth development
- Easier transition to solid food

Nutrition in Childhood

- Appetite becomes unpredictable
- Like familiar foods
- Need high-quality diet

Social environment influences food choices

- Imitate admired people
- Repeated exposure to foods
- Emotional climate, parental pressure
- Poverty

Nutrition Problems

- Little focus on eating
- Too few meals with family
- Not enough fruits and vegetables
- Too many fried foods and soft drinks
- Poverty and lack of nutritional food

Nutrition in Adolescence

- Food intake increases.
- Poor food choices.

Fewer fruits, vegetables, milk, breakfasts.

More soda and high-fat fast food

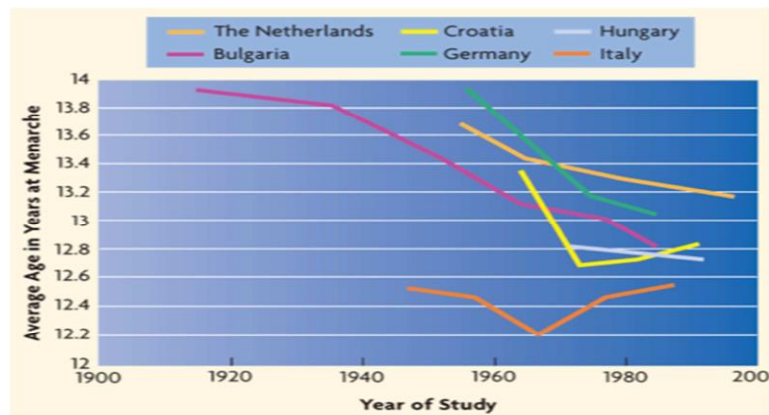
- Iron, calcium, magnesium, vitamin B₂ deficiencies are common.
- More family meals are linked to healthier eating.

Malnutrition:

Types:	Consequences
Marasmus	Physical growth
Kwashiorkor	Growth and weight problems
Iron deficiency	Poor motor development

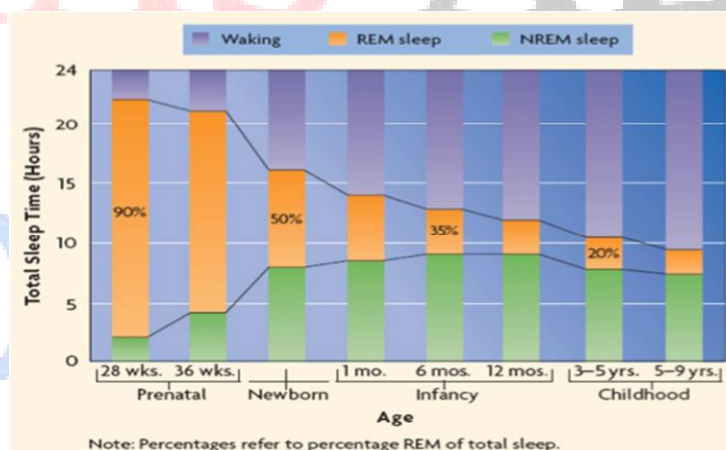
Anemia	Learning. Attention problems
Food insecurity	Passivity, irritability, anxiety

Trends in Menarche:



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Childhood Sleep Changes:



Note: Percentages refer to percentage REM of total sleep.

Summary:

We looked at major nutritional needs of the child and associated physical growth. We also looked at sexual maturation. Finally we looked at sleep patterns of the growing individual.

LESSON 21

BRAIN, BODY, MOTOR SKILLS AND SEXUAL MATURATION

Being over or underweight and Obesity

Overview:

We will discuss being overweight and becoming obese.

We will discuss factors that lead to being overweight and obese, and the effects of being fat.

Finally we will take a look at being underweight and diseases associated with being underweight.

Critical Questions:

What are the causes and risks of being overweight and obese?

How are some developing nations moving towards these problems? What are some of the diseases associated with being underweight?

Definition:

Being overweight or being obese means an abnormal accumulation of body fat, usually 20% or more over individual ideal weight. We use body mass index (BMI) to measure to access weight issues.

Weight	BMI
Underweight	< 18.5
Normal	18.6 - 24.9
Overweight	25.0 – 29.9
Mildly Obese	30.0 – 34.9
Moderately Obese	35.0 – 39.9
Extremely Obese	> 40.0

$$\text{BMI} = \text{Weight (kg)} / \text{Height (m)}^2$$

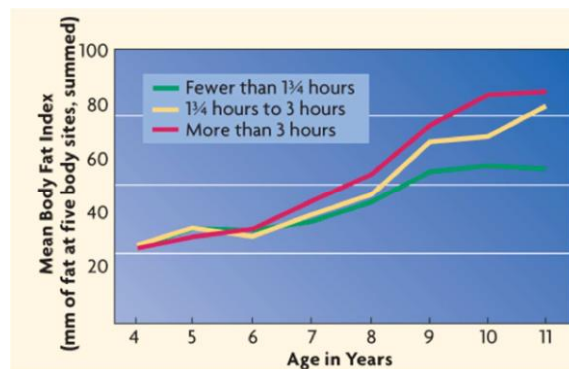
Categories:



Causes:

- Genetics (Overweight parents)
- Gender (Women are more susceptible)
- Activity (Reduced; Television).
- Socioeconomic Status (Poverty)

- Alcohol
- Smoking (Cessation)
- Prescribed Drugs (Tricyclics)
- Family and cultural eating habits
- TV Watching

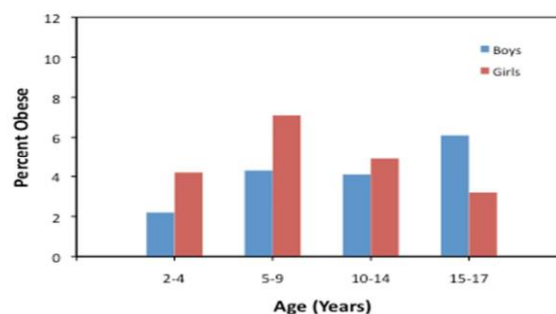


Cheeseburger

Size of the cheeseburger today has increased in calories (from 333 to 590 calories) over a 20-year period.

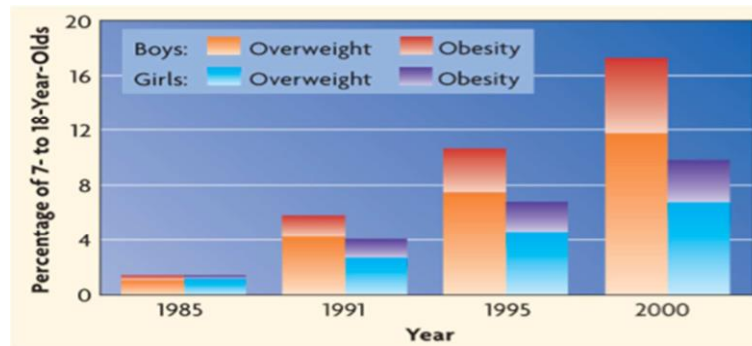
Obesity Across Gender:

The data below comes from Australia and depicts general trends in obesity in boys and girls across four age groups.



Overweight and Obese: China

The following data depicts overweight and obese children (7-18 years) in china. Boys tend to show greater obesity than girls.

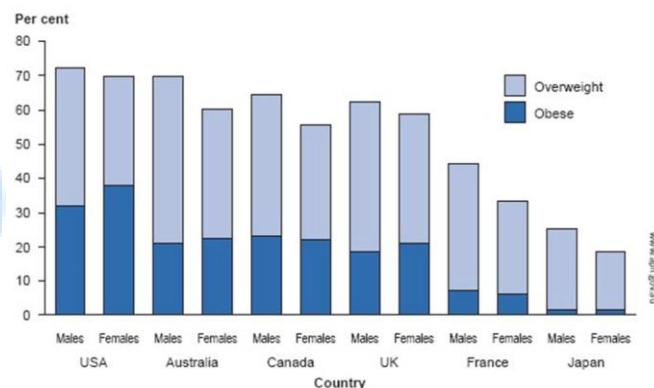


Number of Obese (US)

The total number of obese in the US has steadily increased from 1980 to present times. It is expected to reach 45% by the year 2020.

Nation Comparison

See below for a comparison of being overweight or obese among many different nations of the world.



Health Risks

- Blood pressure, cholesterol
- Respiratory problems
- Diabetes
- Liver, gall bladder disease
- Sleep, digestive disorders
- Cancer
- Early death

Psychological & Social Consequences:

- Feeling unattractive

- Stereotyping
- Teasing, social isolation
- Depression
- Problem behaviors
- Less schooling, lower income, marriage problems

Underweight:

By definition an underweight individual is below 18.5 BMI.

Underweight Problems:

- A number of physical and mental diseases are associated with being underweight, e.g., hyperthyroidism, cancer, and tuberculosis.
- Women suffer from amenorrhea and osteoporosis.
- Anorexia nervosa.
- Increased mortality rate for the obese individuals

Anorexia Nervosa:

- An eating disorder in which the individual is afraid of gaining weight. Anorexia begins early in life around 9 -14 years and continues as the individual grows. It occurs in girls 10 times more than boys and can affect individuals of any race, SES or cultural background.
- Anorexia is never truly cured but to a great extent managed by medical assistance.
- Individuals with anorexia usually have a distorted body image.

Summary:

We looked at what is meant by being overweight and becoming obese. We also discussed factors that lead to being overweight and obese, and the effects of being fat. Finally we looked at being underweight and diseases associated with being underweight.

LESSON 22

BRAIN, BODY, MOTOR SKILLS AND SEXUAL MATURATION

Need for Exercise, Play for growth and Security

Overview

We will look at the effect of exercise on growth and development of the child.

We will also study need for play during development.

And finally look at sense of security as the child grows into adulthood.

Critical Questions

Why does the growing child need exercise?

Why is play and games important during development?

And how are exercise, play and other factors lead to security in the individual?

Exercise

Exercise is any bodily activity that enhances physical fitness, health and wellbeing. Exercise during growth and development; promotes strength, improves cardiovascular, develops athletic skills and maintains appropriate body weight.

Kinds of Exercise

Exercise kinds are three:

1. Flexibility exercise
2. Aerobic exercise
3. Anaerobic exercise

Flexibility exercise largely consists of stretching.

Aerobic exercise includes, walking, running, hiking, swimming, tennis etc.

Anaerobic or resistance exercise consists of weight training, functional training, and sprinting.

Though all kinds of exercises are recommended for children, and adolescents, however the intensity of exercise needs to be lower for younger children.

Exercise and Disease

Exercise improves physical skills and motor coordination during growth and development.

During development exercise can boost the immune system in the child.

This can prevent diseases like cardiovascular disease, type-2 diabetes and obesity.

Exercise can also prevent mental diseases like depression and maintain positive mental outlook on life.

Kinds of Play

1. Exploratory Play (rattles, squeeze toys).
2. Construction Play (blocks, model kits).
3. Pretend & Role Play (dolls, stuffed toys, cars).
4. Game & Activity Play (jigsaws, video games)
5. Sports & Recreational Play (tricycles, bicycles)
6. Media Play (Paint, paper, musical instruments).
7. Educational & Academic Play (books, educational software).

Exploratory Play

Rattlers and squeeze toys are used at an early age in the child's life (infancy). These toys make the infant learn elementary motor skills, interact with the environment and get better control over their senses.



Construction Play

At a slightly more advanced stage in infancy, the child can engage in construction play. In this play the infant or a growing child uses blocks and model kits to engage in creativity.



Pretend & Role Play

From a very young age the children start engaging in role or pretend playing. They become mommies and "doctors" to their dolls and play objects. Children learn roll playing, social interactions.



Game & Activity Play

Young children play games like jigsaw puzzles or video games to improve motor skills, eye hand coordination and interaction with others.



Sports & Recreational Play:

When children become older and a little stronger and are able to walk they start playing recreational games. Such play boosts their self-esteem and pleasure in achievement.



Media Play:

Painting, playing with musical instruments constitutes media play. Children learn and become creative in these play behaviors.



Educational & Academic Play:

Painting, playing with musical instruments constitutes media play. Children learn and become creative in these play behaviors.



Security:

Early in life children get their security from their parents and caregivers and then objects they find pleasure in. Soft cuddly blankets and pillows provide sense of security.

However when children become older and are more independent, they develop a sense of security about themselves.

Playing games and doing exercise requires them to become aware of their security issues.

Parental Security & Safeguards:

Even though relatively older children have a sense of security while playing or exercising, they still need parental supervision.

Parents need to know “where”, “what” and “who” the child is with.

Children learn to close doors, windows etc. when home alone, developing a sense of security.

Around adolescent age children have a sense of security that is similar to adults.

Summary:

We looked at the effect of exercise on growth and development of the child. We also studied need for play during development. And finally look at sense of security as the child grows into adulthood.

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